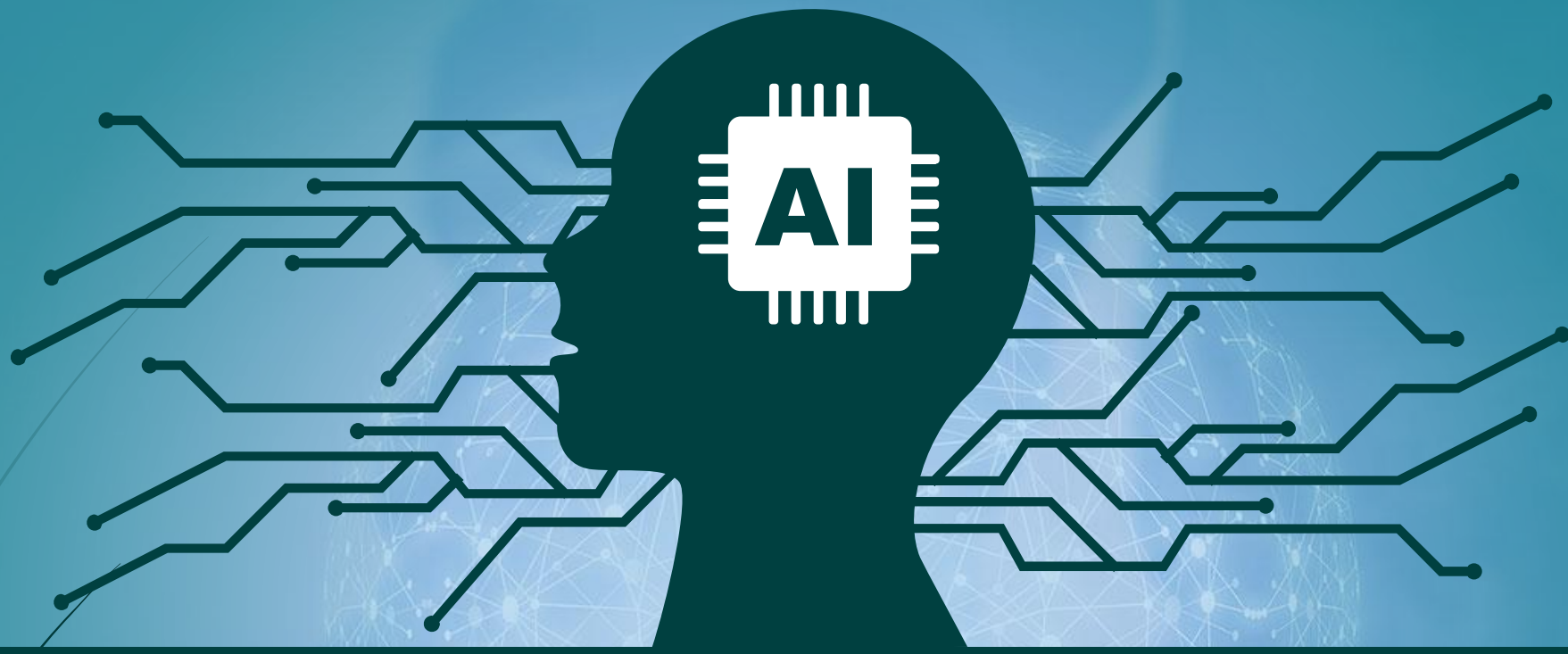


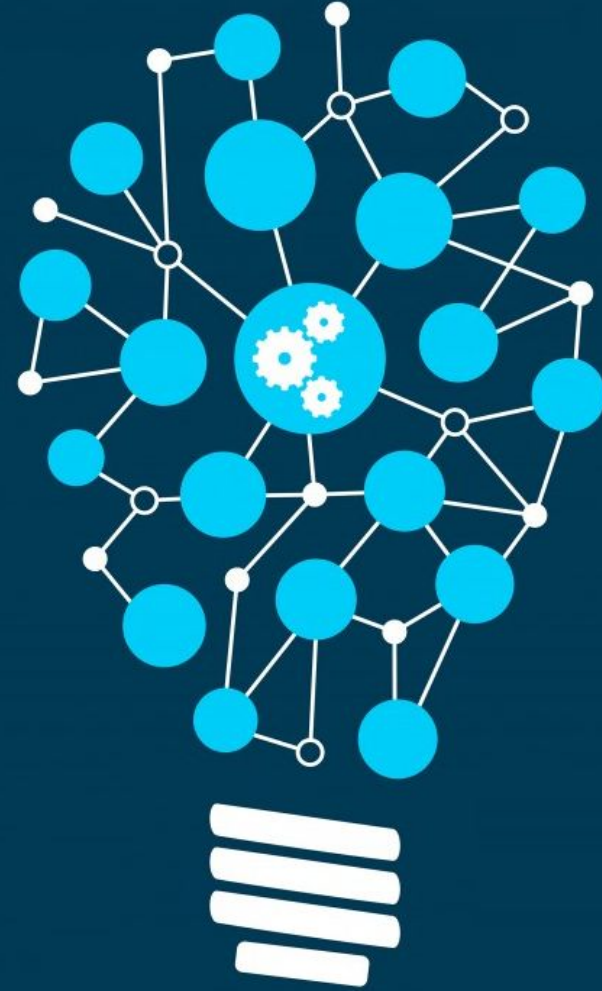


Advanced Artificial Intelligence

Week #6

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MACHINE LEARNING

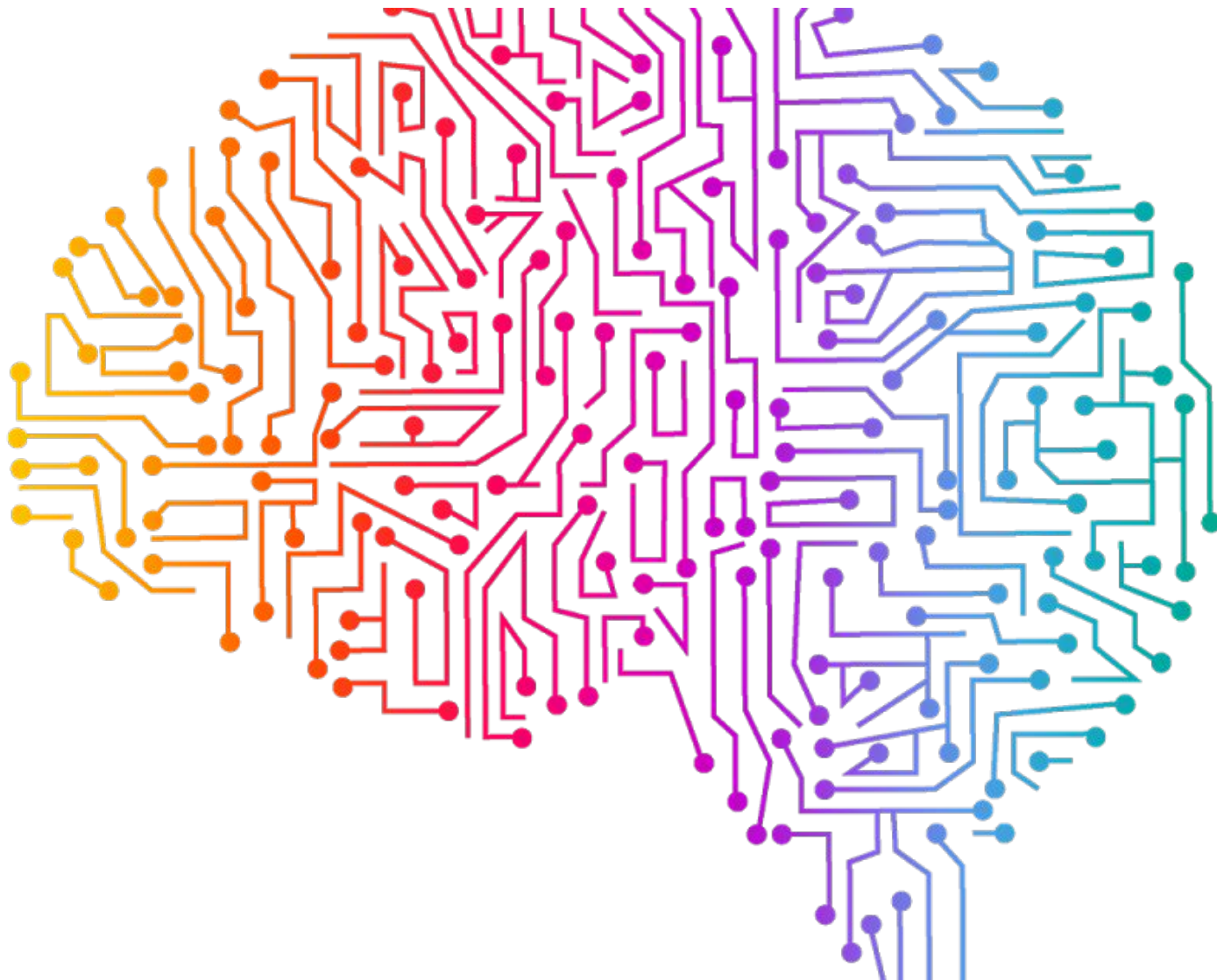




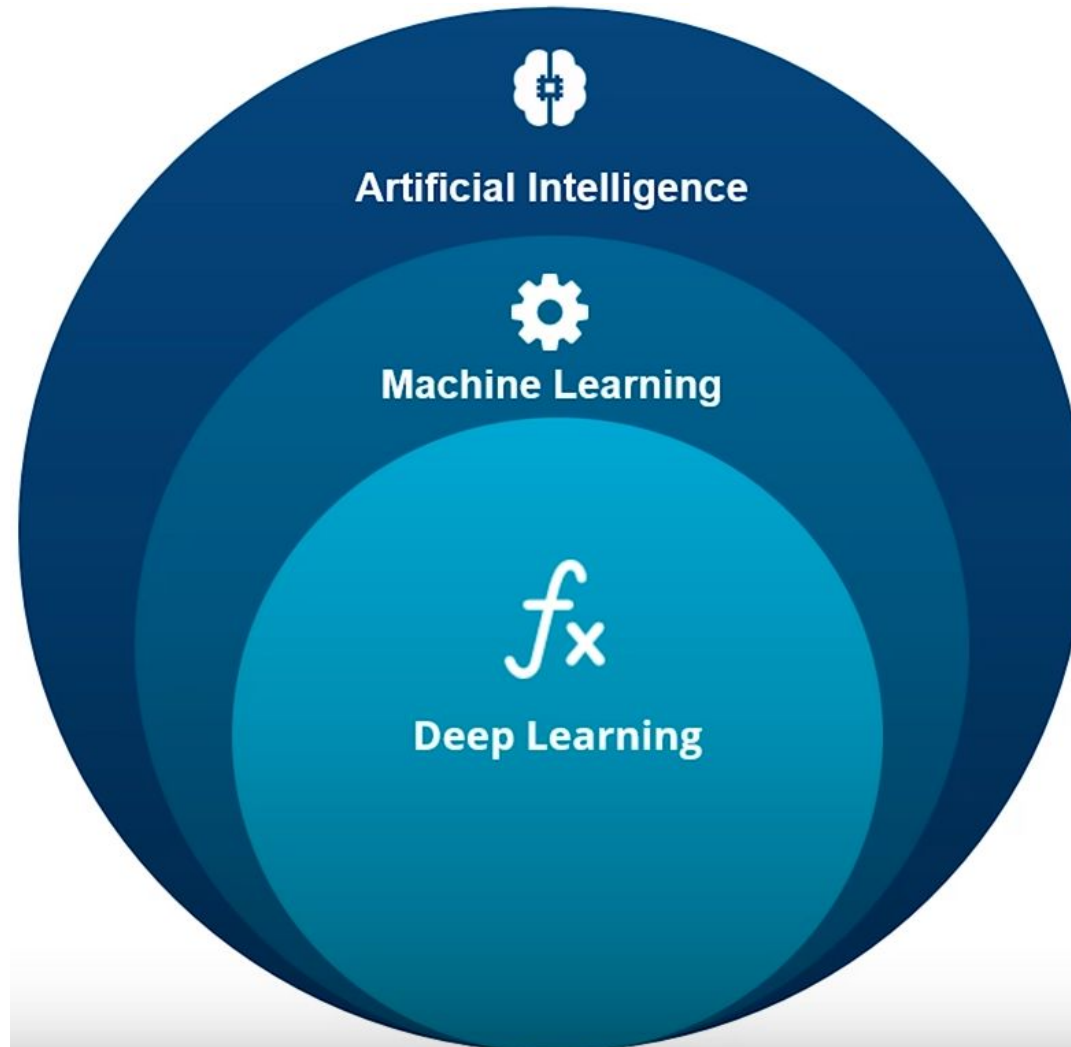
Learning Objective of this Topic

- AI, ML, DL (Difference)
- What is Machine Learning?
- Types of Machine Learning
- Supervised Learning
 - K-Nearest Neighbour Classification Algorithm
 - Examples
 - Real World Applications
 - Advantages & Disadvantages
 - Class Activity

AI, ML, DL
(Difference)



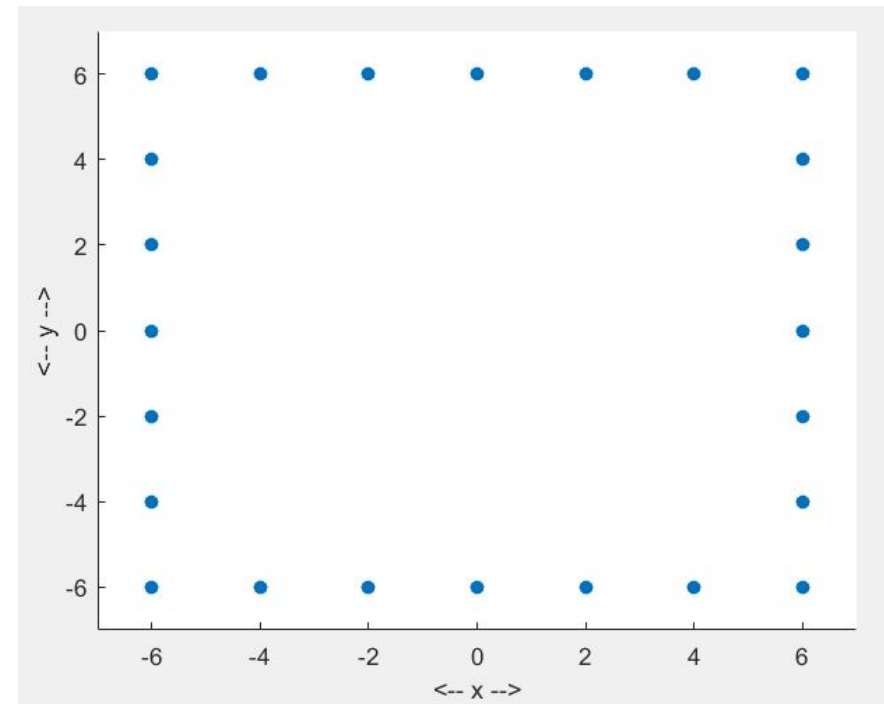
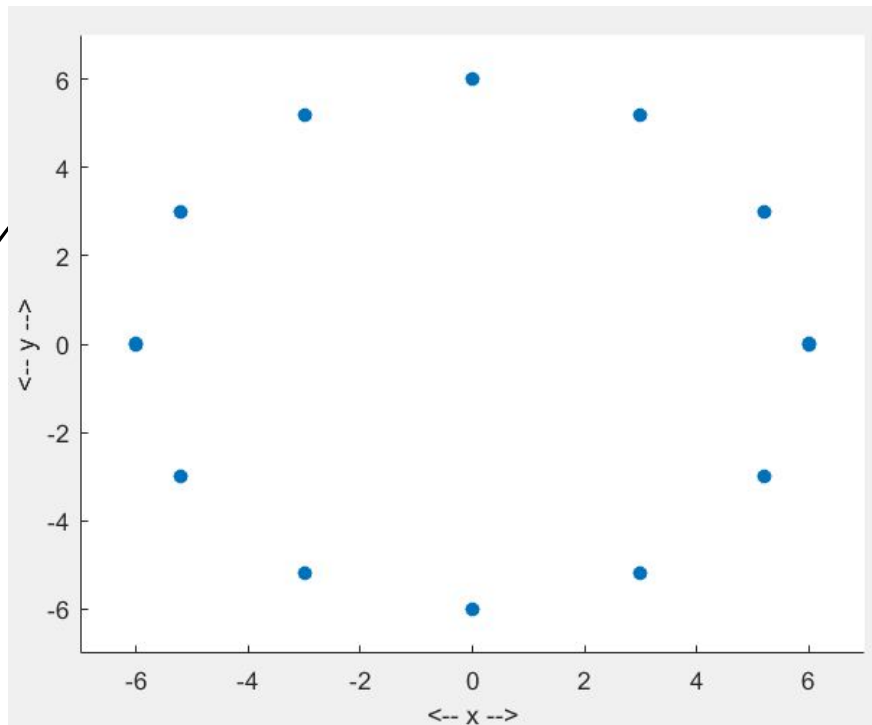
AI vs Machine Learning vs Deep Learning



Using AI

Problem:

□ Given a set of connected points, find if the points form a **circle** or a **square**?



Using AI

□ Features

- 1) Number of corners
- 2) Number of equal Sides

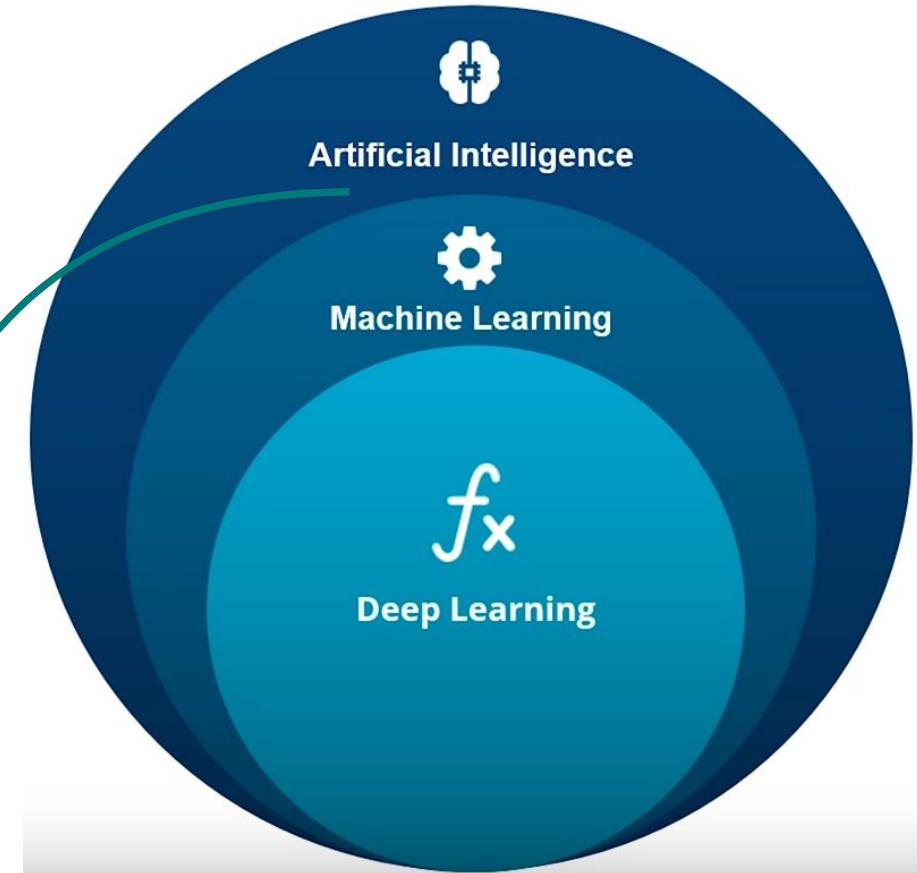
Rule base:

If (number of corners == 4) and (equal sides == 4)

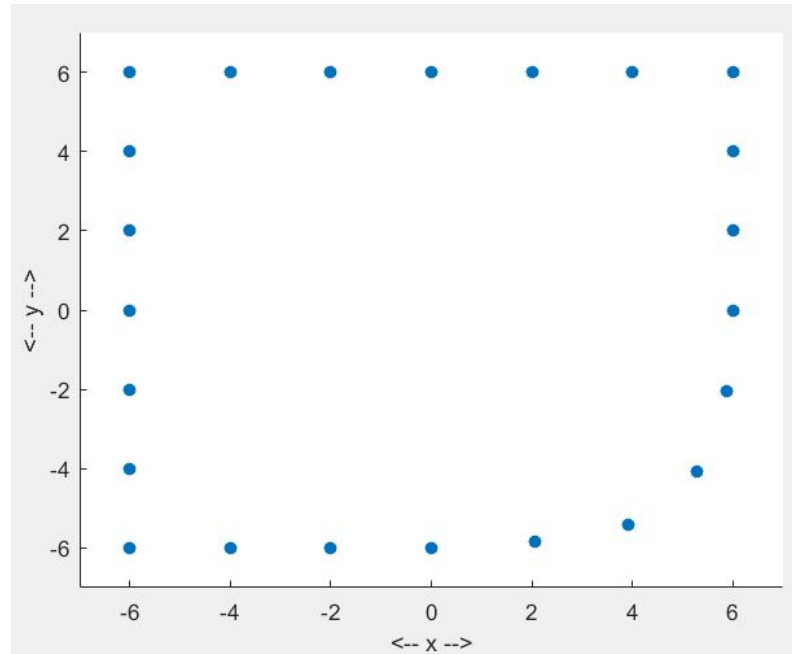
points form square

Else

points form circle



Problem with classical AI



Square or a circle?

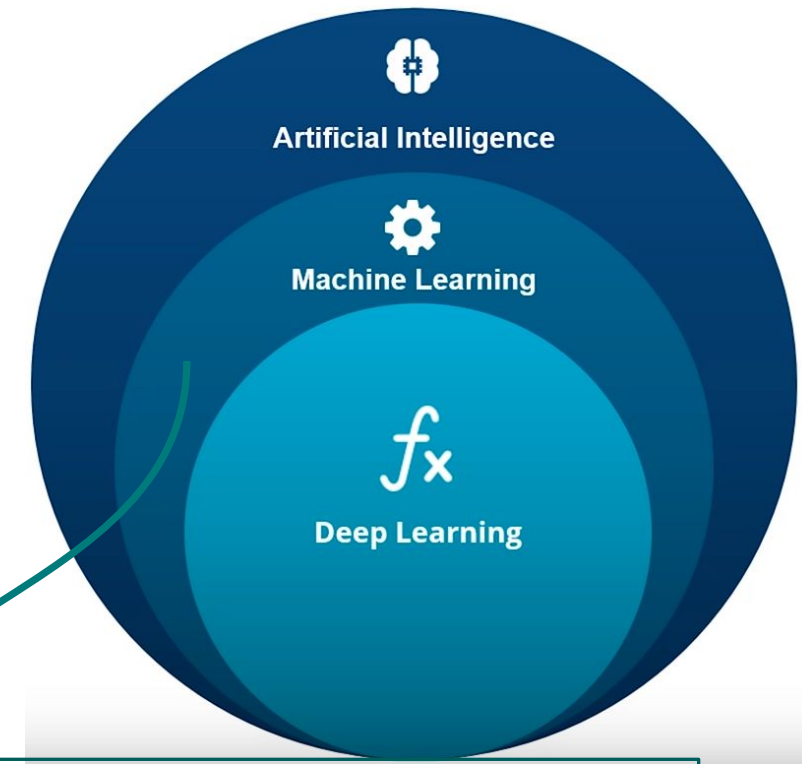
Using ML

□ Features

- 1) Number of corners
- 2) Equal Sides

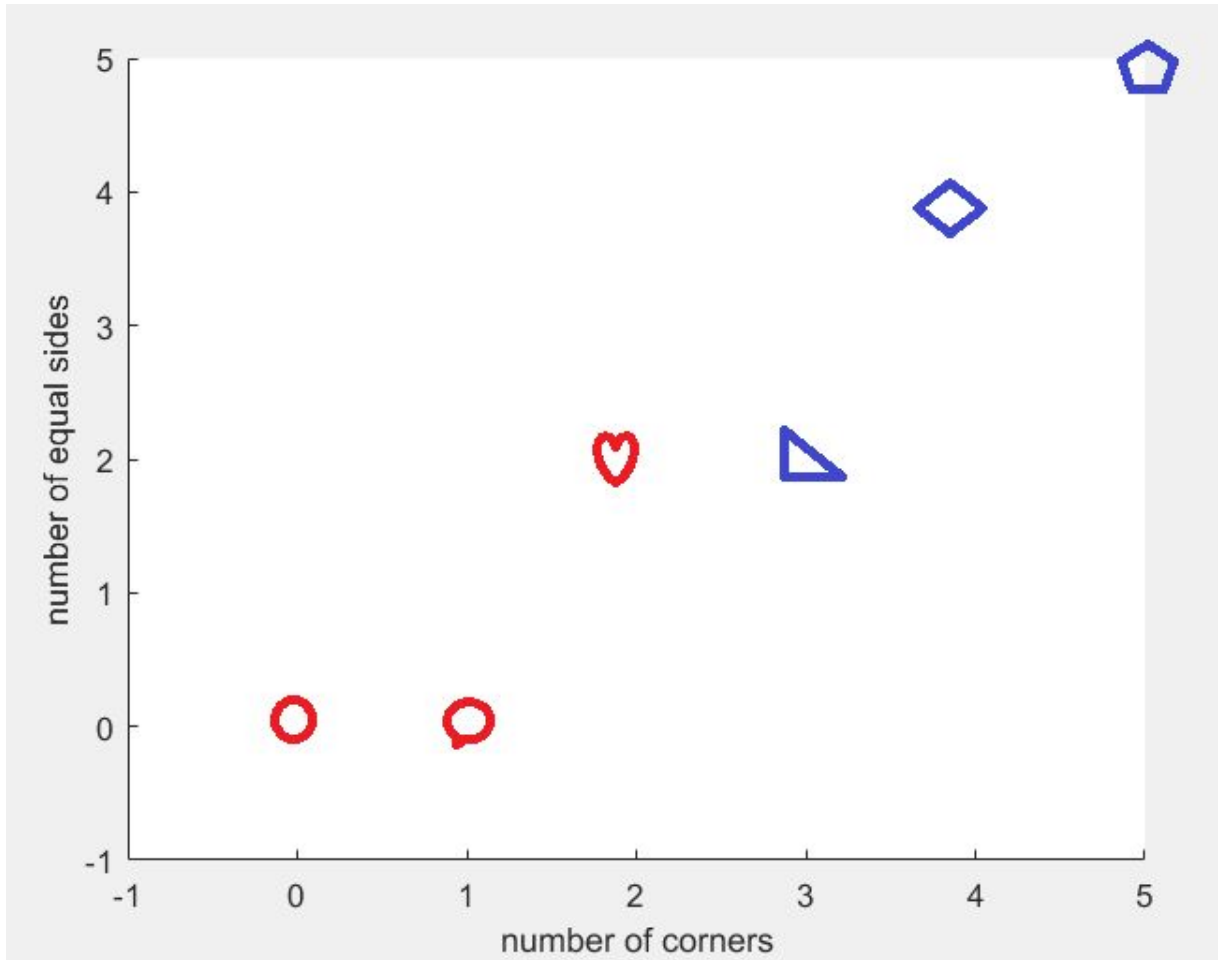
Training Phase:

- 1) Take 3 examples of circles and 3 examples of squares
- 2) For each example, compute features (number of corners, number of sides that are equal)
- 3) Also tell the computer that whether extracted features are taken from a circle or a square



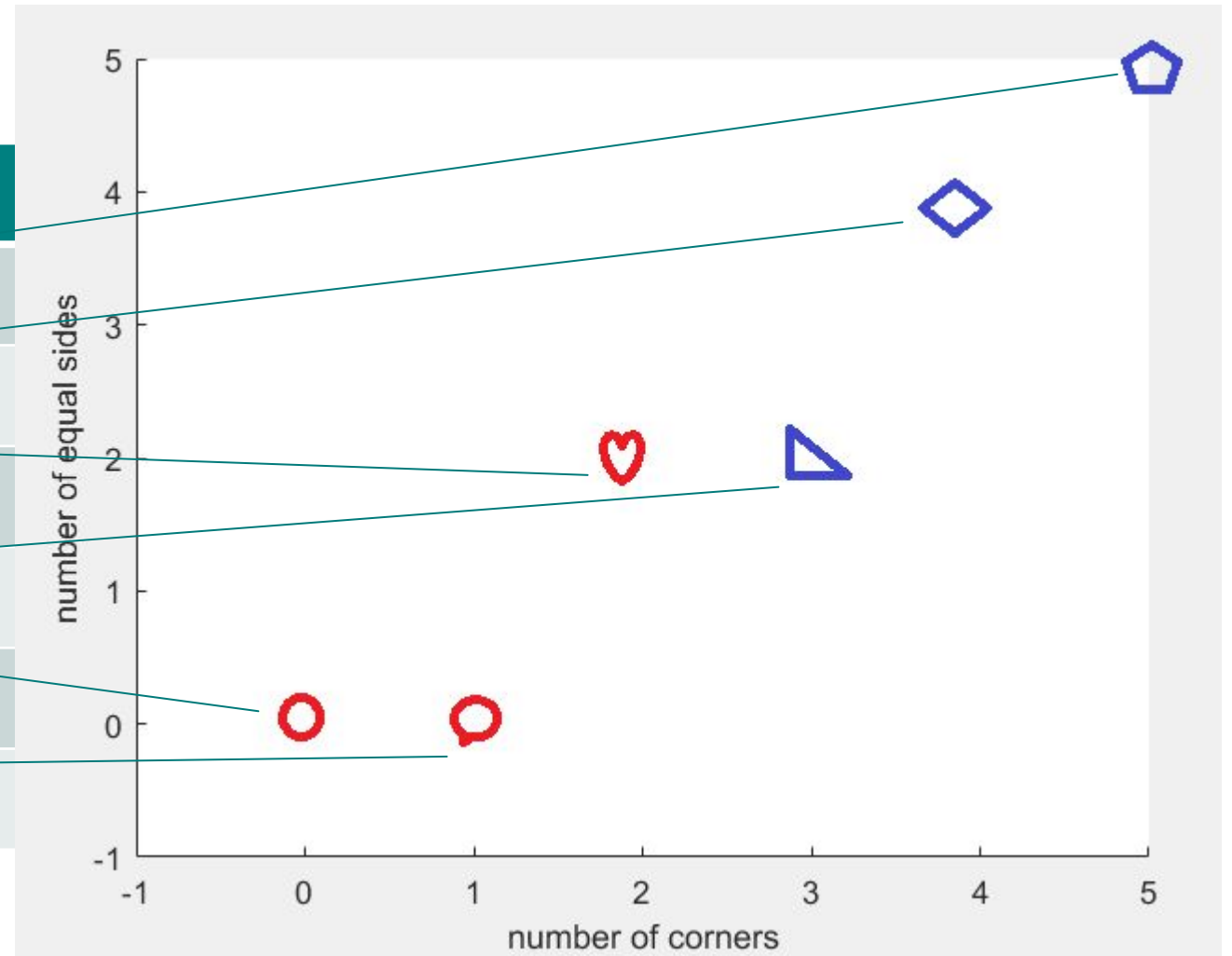
Using ML: Feature space

- During training, we give machine all examples of different shapes that look similar to a circle or square
- Red ones similar to a circle, blue ones similar to a square.



Using ML: Feature space

Class
Square
Square
Circle
Square
Circle
Circle



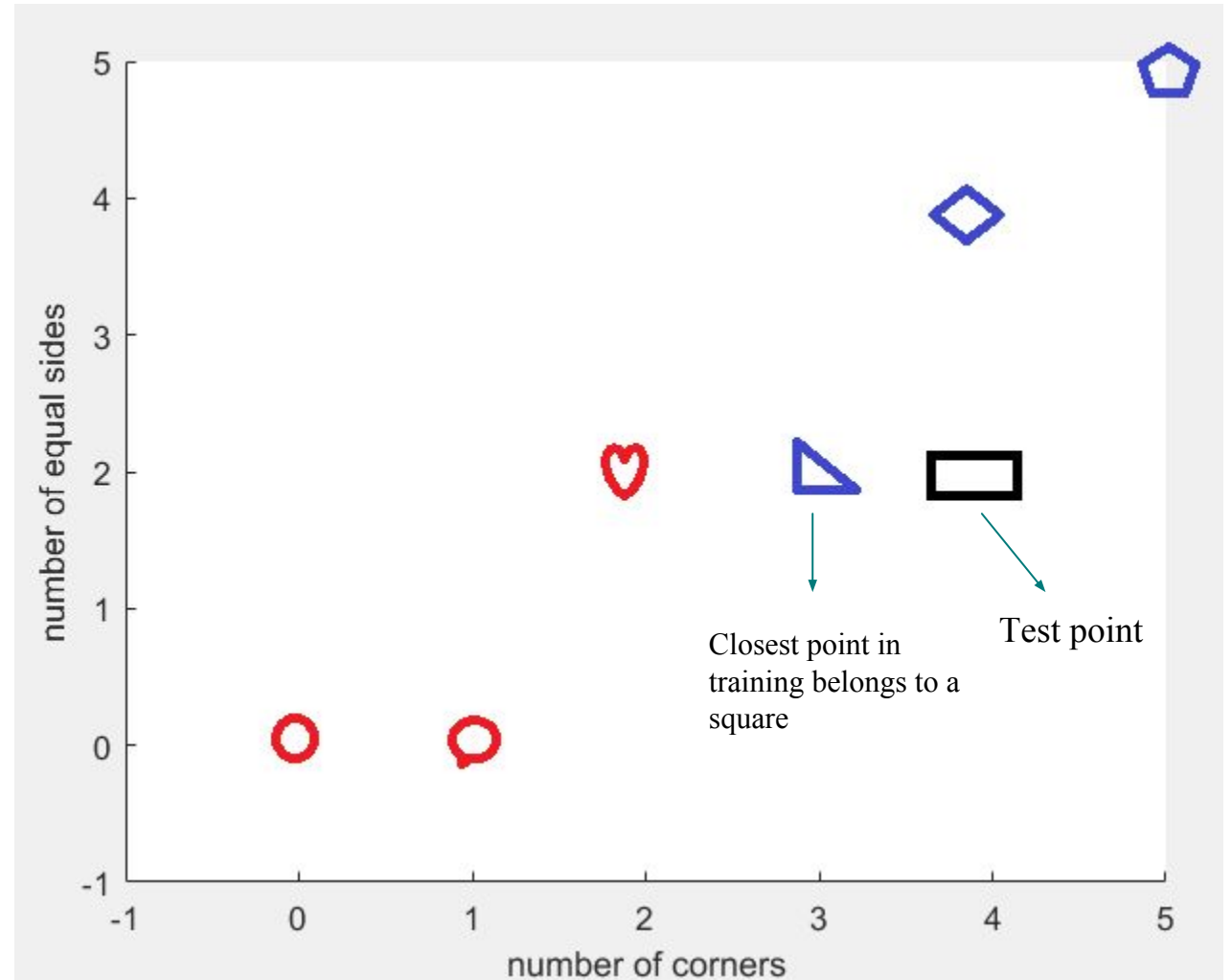
Using ML: Feature space

Testing Phase:

Let's determine whether a set of points belonging to a rectangle is classified as a square or a circle?

Nearest Neighbor Approach:

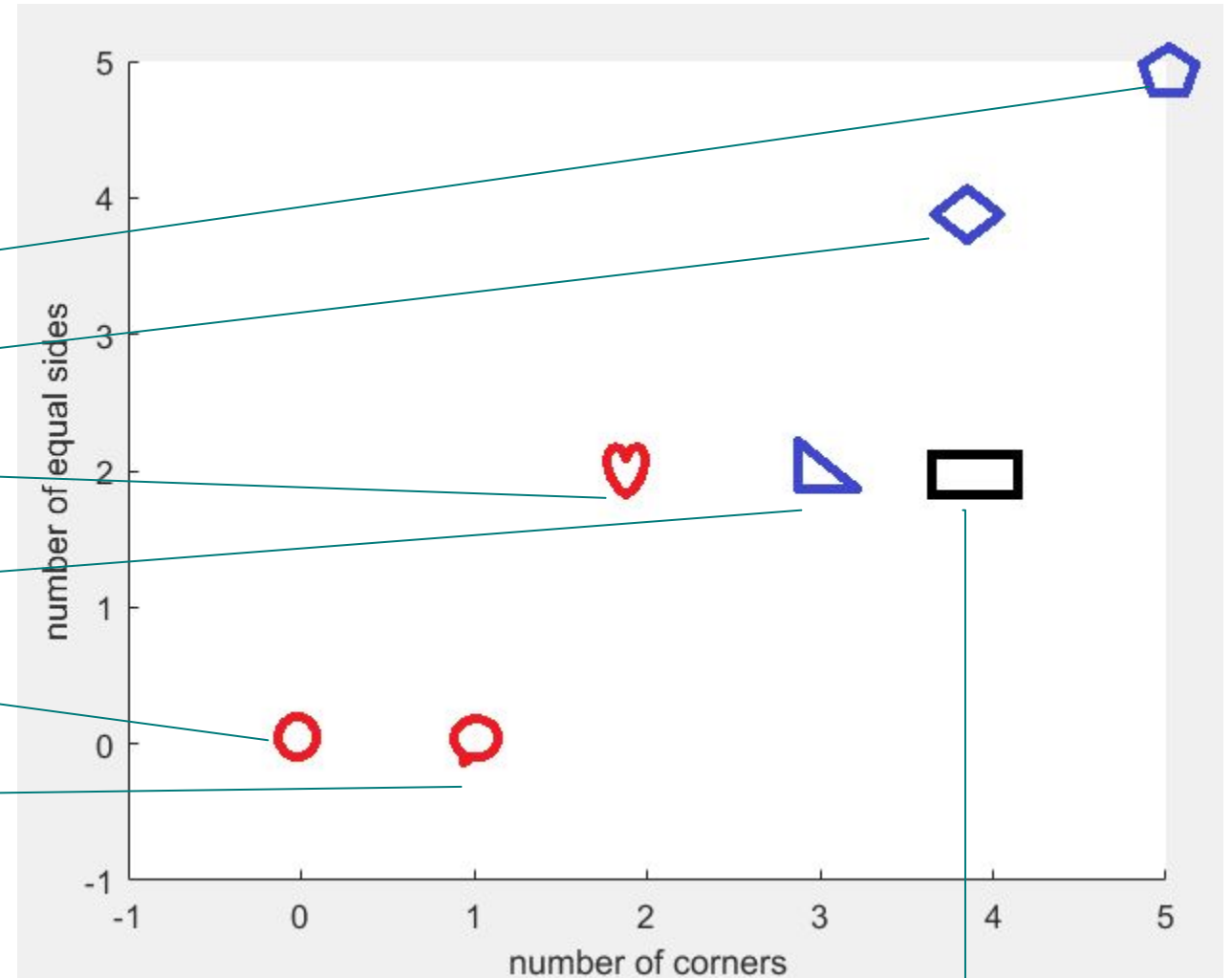
- Compute features of the test object and find that example in training the features of which is most similar to this test case.



Using ML: Feature space

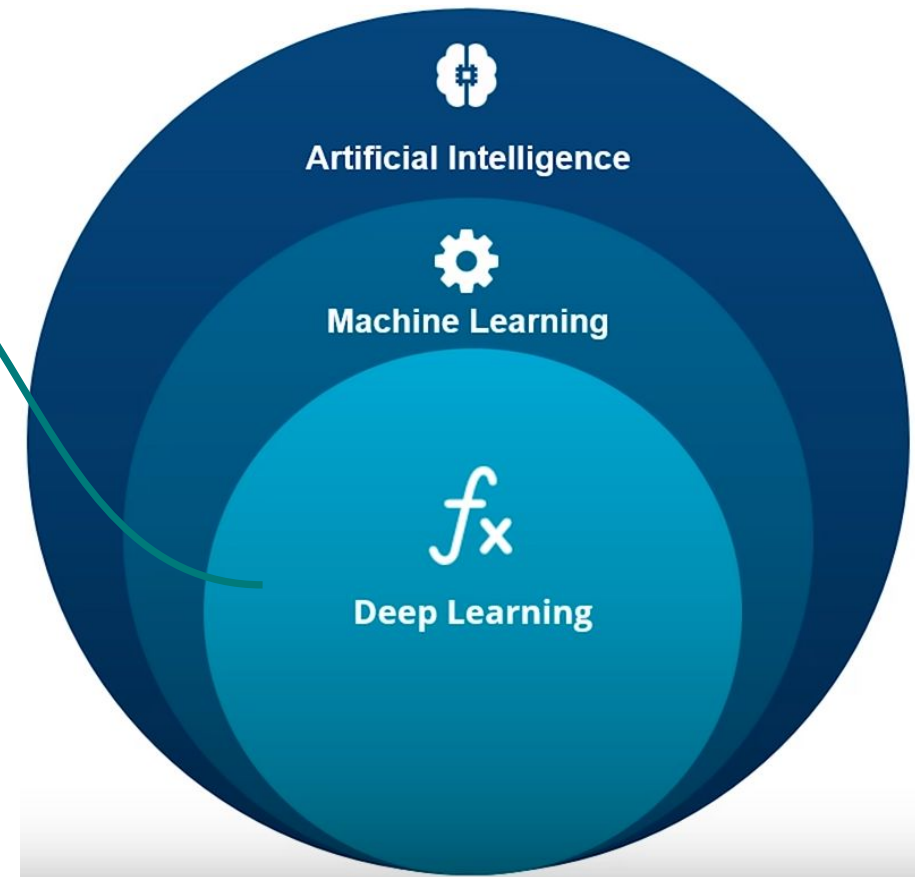
Class	Number of corners	Number of equal sides
Square	5	5
Square	4	4
Circle	2	2
Square	3	2
Circle	0	0
Circle	1	0

Class	Number of corners	Number of equal sides
?	4	2



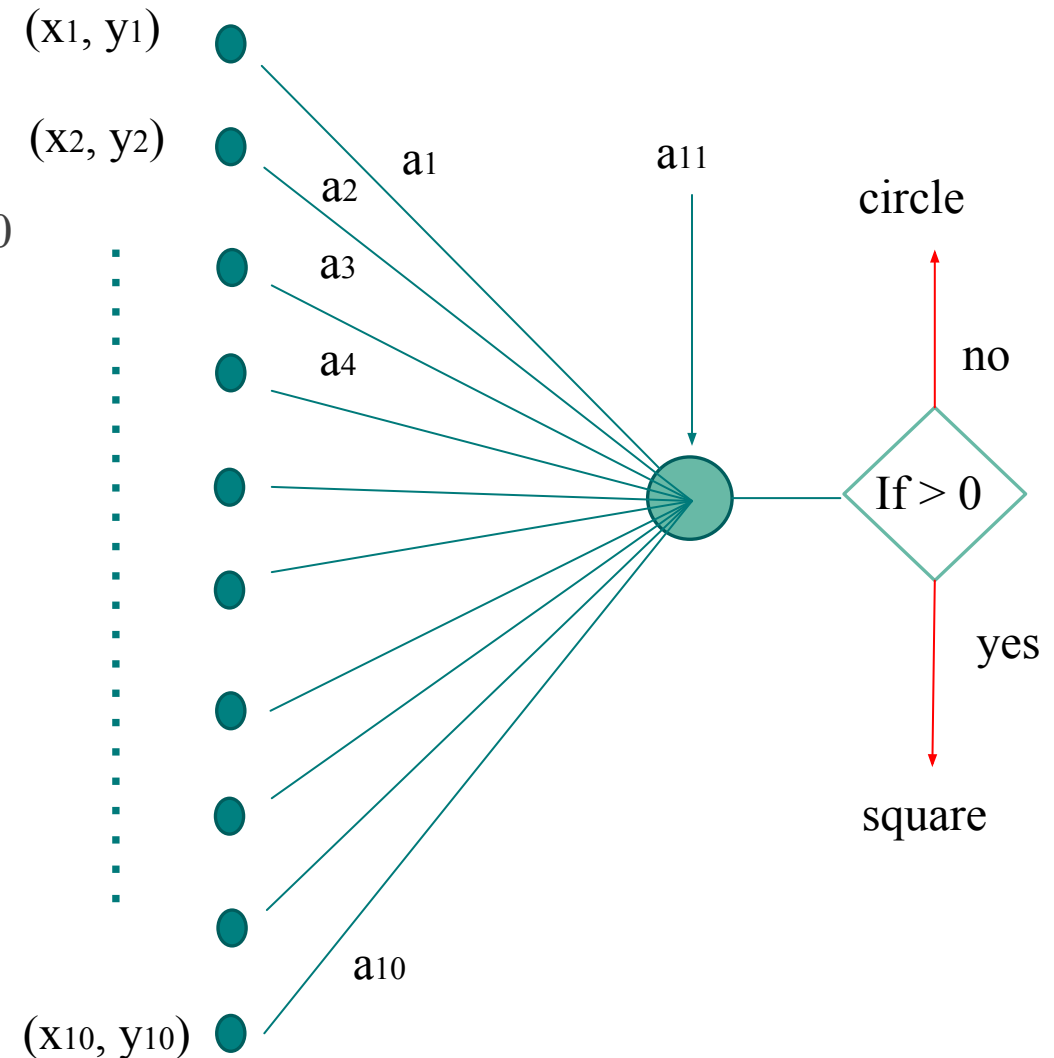
Using DL

- ❑ No need of determining features
- ❑ Just let an artificial neural network learn what type of features are useful for a given task.
- ❑ In classical AI and ML, we don't feed all points to an algorithm directly. We first compute features and then feed those computed features only.
- ❑ In DL, we don't compute features, we feed input directly to a network.



Using DL

- Assume that every example contain 10 points arranged in circle or a square.
- Training Phase:
 - 1) Take 3 examples of circles and 3 examples of squares
 - 2) Also tell the computer that whether extracted features are taken from a circle or a square
 - 3) For each example, adjust 11 parameters such that the output for each example is +ve for circle and -ve for square.



$$a_1*(x_1+y_1)+a_2*(x_2+y_2)+\dots+a_{10}*(x_{10}+y_{10})+a_{11} > 0 \text{ (for all square examples in training)}$$

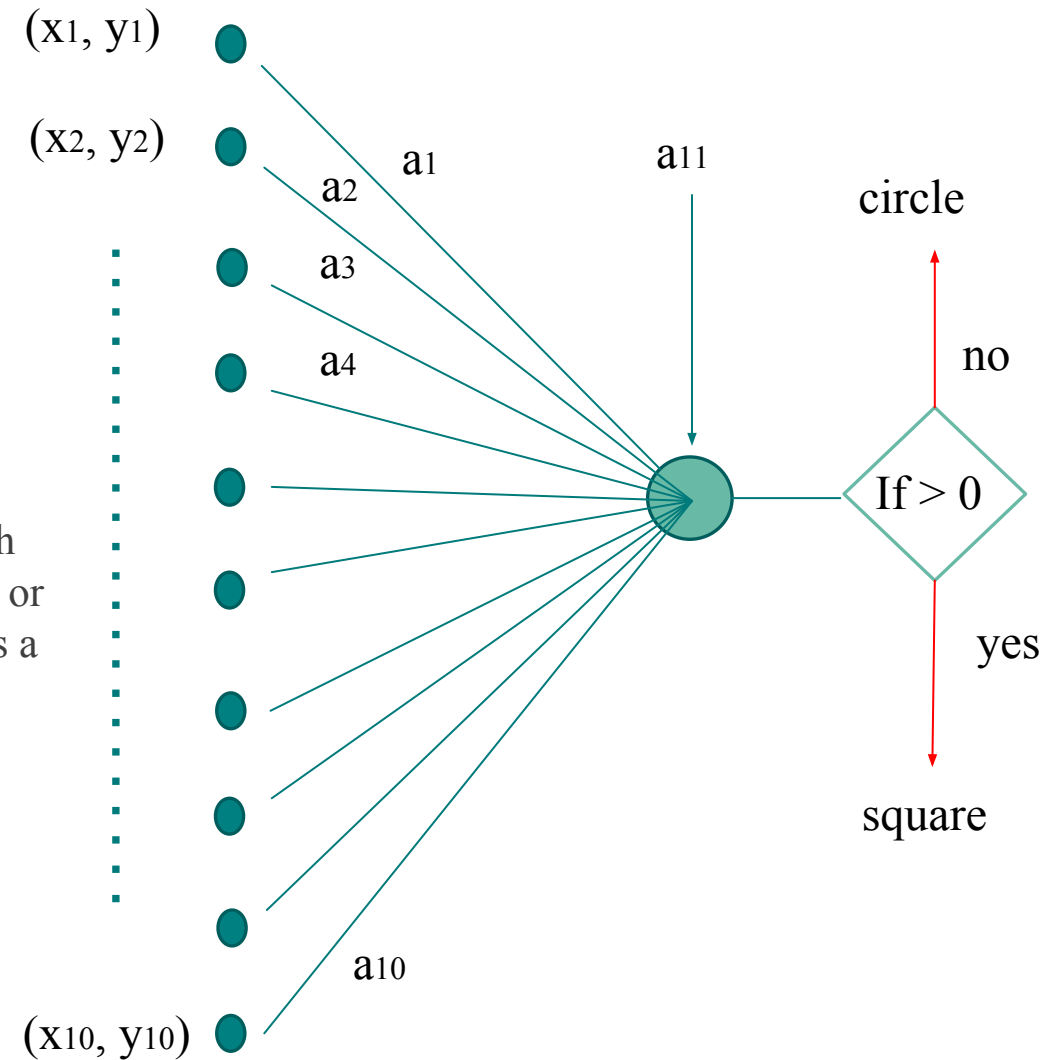
$$a_1*(x_1+y_1)+a_2*(x_2+y_2)+\dots+a_{10}*(x_{10}+y_{10})+a_{11} \leq 0 \text{ (for all circle examples in training)}$$

Using DL

- The whole purpose of training is to calculate set of 11 parameters ($a_1, a_2, \dots, a_{11}$)

- Test Phase:

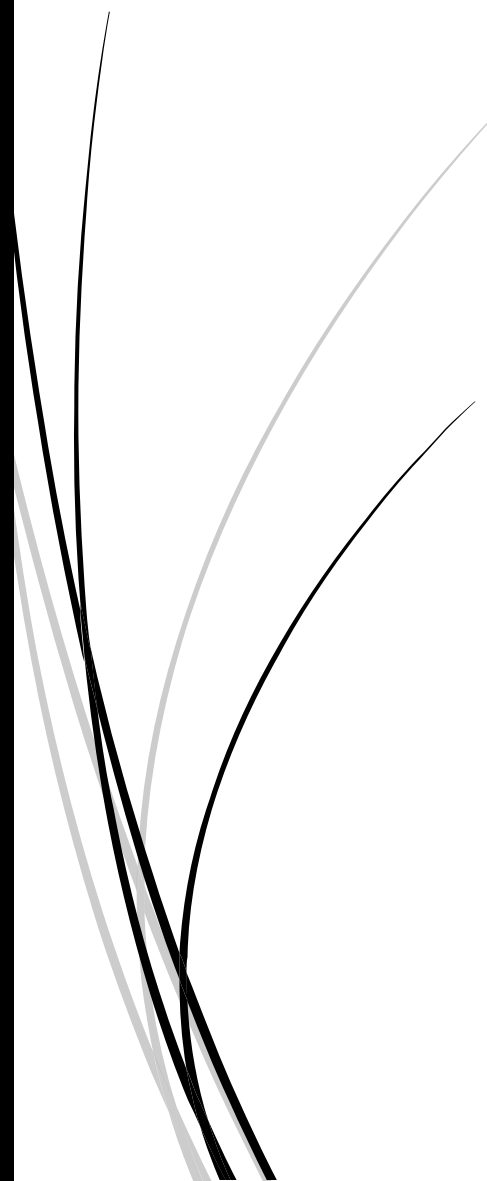
Multiply and add those 11 parameters with input test points to see if the output is +ve or -ve. If +ve then given test case/example is a square, otherwise a circle.



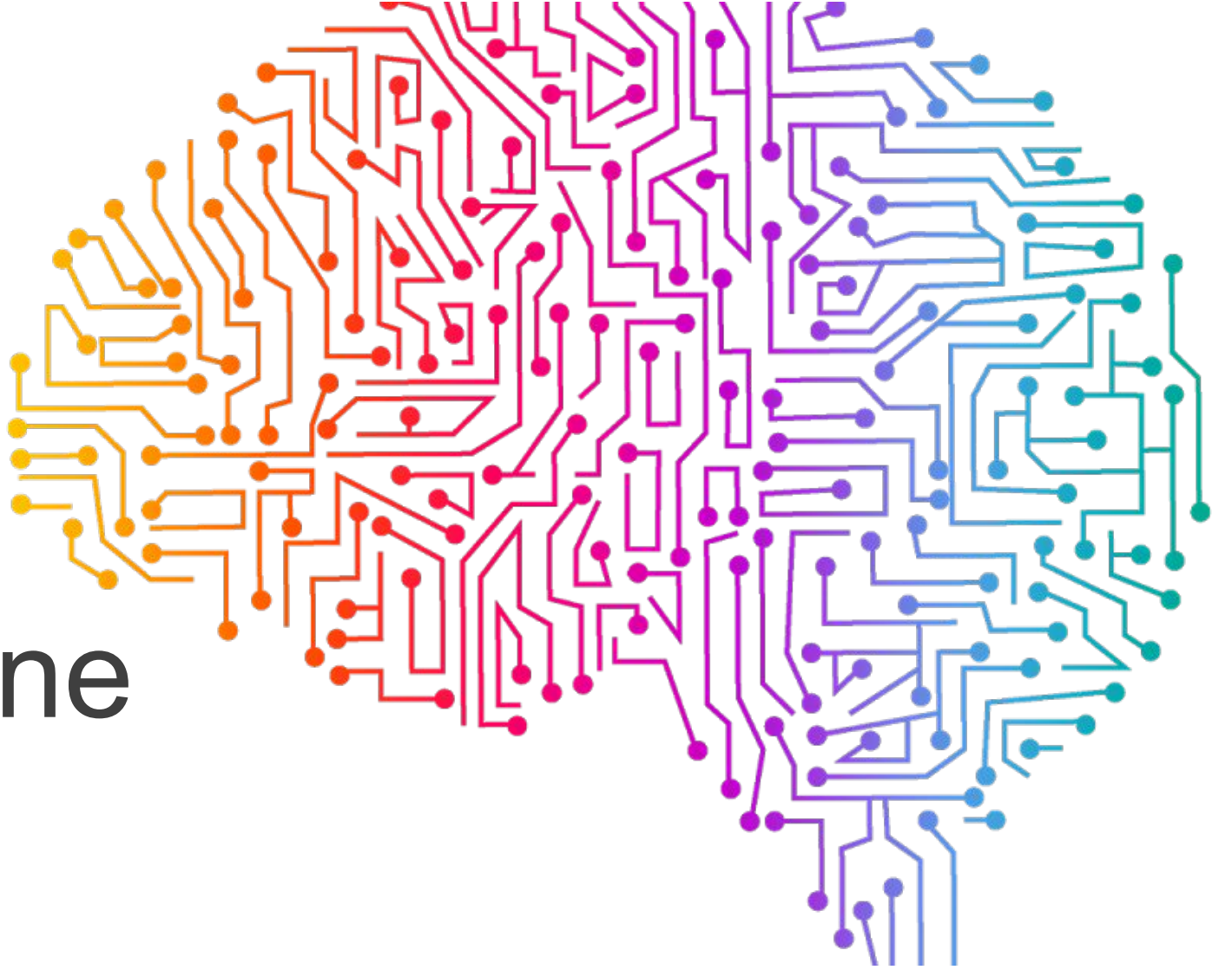
$$a_1*(x_1+y_1)+a_2*(x_2+y_2)+\dots+a_{10}*(x_{10}+y_{10})+a_{11} = ?$$



Summary

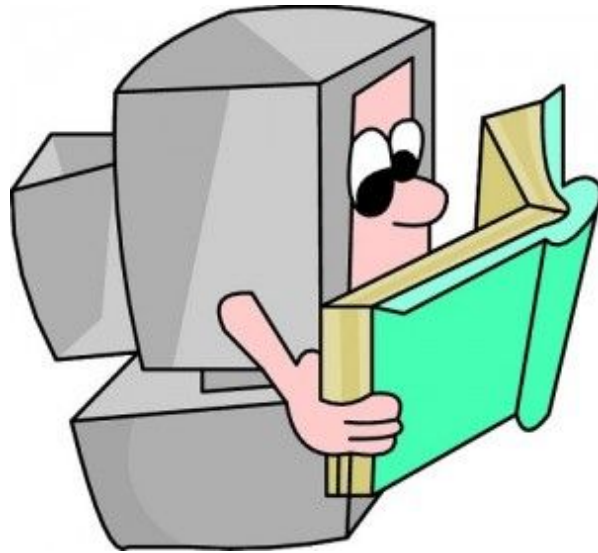


Scheme	Training Set required	Pre-Determined Features required
AI	No	Yes
ML	Yes	Yes
DL	Yes	No

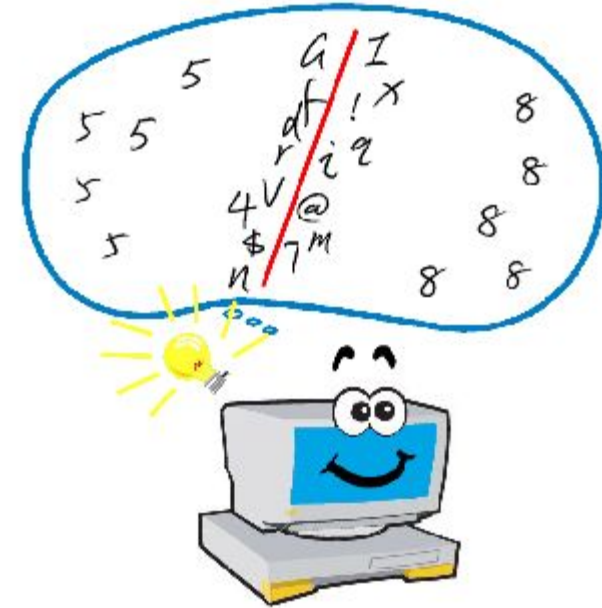


What is Machine
Learning?

What is Machine Learning?



Make the machine
'learn' some thing



Evaluate how good the
machine has '*learned*'

Machine Learning

“Machine learning is concerned with computer programs that automatically improve their performance through experience.”

- Herbert Simon



“A computer program is said to learn from **experience E** with respect to some **classes of tasks T** and **performance measure P**, if its performance at task in T, as measured by P, improves with the experience E.” (Tom Mitchell 1998)

- Learning = Improving with experience at some task
 - Improve over task T,
 - With respect to performance measure, P
 - Based on experience, E.



Machine Learning

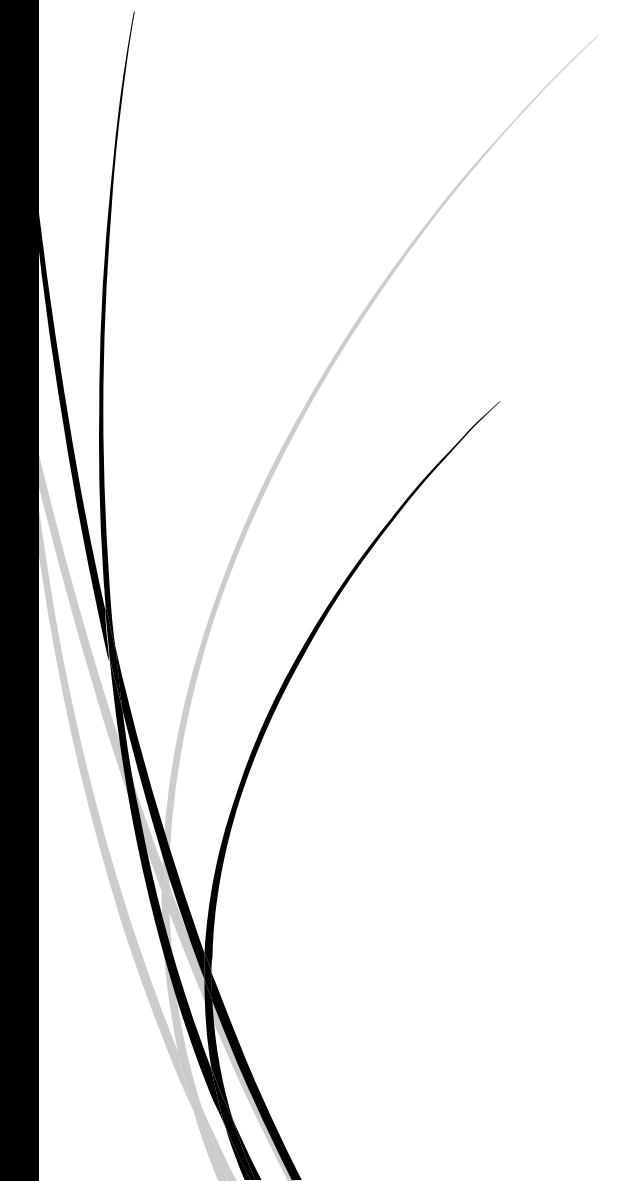


Field of study that gives computers the ability to learn without being explicitly programmed.

Arthur Samuel (1959)



Machine Learning



Machine learning is programming computers to optimize a performance criterion using example data or past experience.

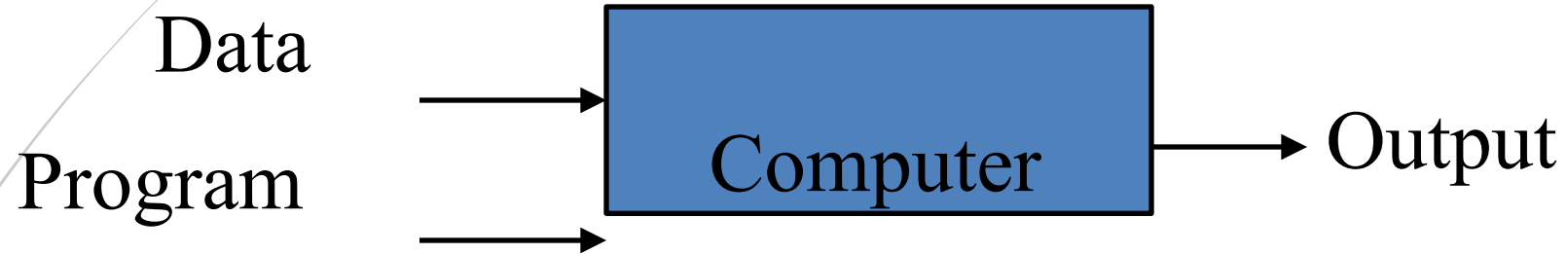
Tom Mitchell (1998)

Machine Learning

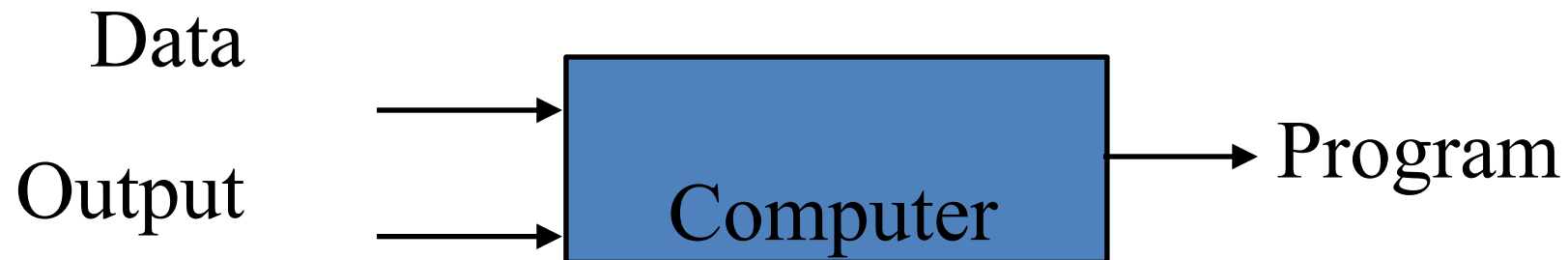
- Learning = Improving with experience over some task

A computer program is said to *learn* from experience E with respect to some task T and performance measure P , if its performance at task T , as measured by P , improves with experience E

Traditional Programming



Machine Learning



Learning Problems – Examples

- Handwriting recognition learning problem
 - **Task T :** recognizing handwritten words within images
 - **Performance measure P :** percent of words correctly recognized
 - **Training experience E :** a database of handwritten words with given classifications



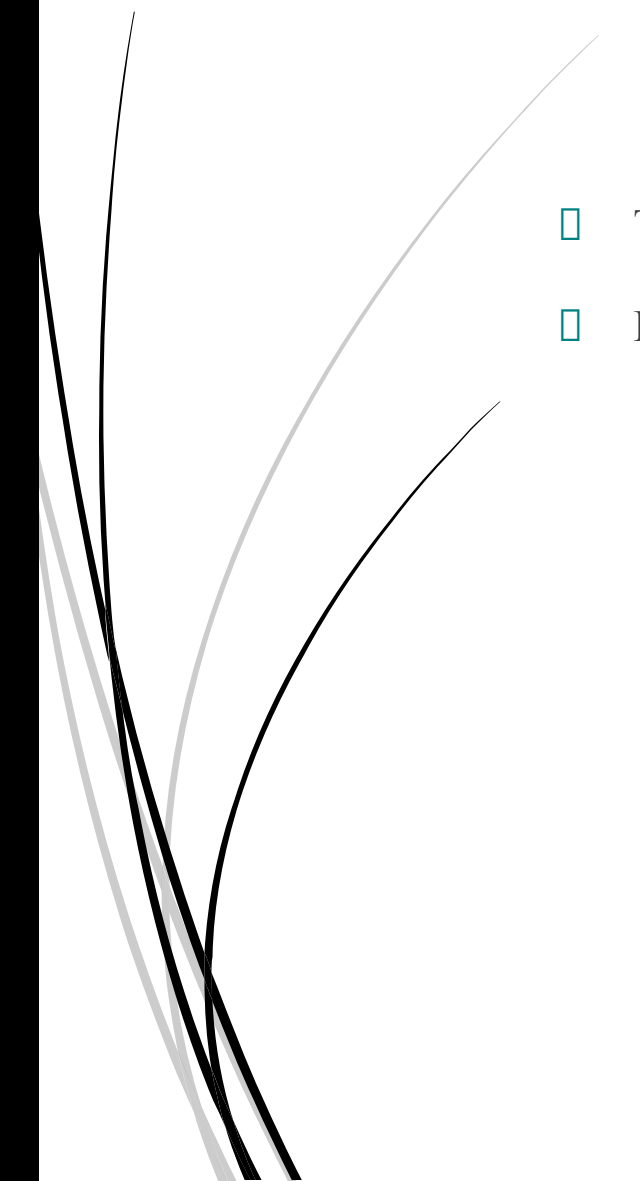
Learning Problems – Examples

- A robot driving learning problem
 - **Task T :** driving on public four-lane highways using vision sensors
 - **Performance measure P :** average distance traveled before an error (as judged by human overseer)
 - **Training experience E :** a sequence of images and steering commands recorded while observing a human driver



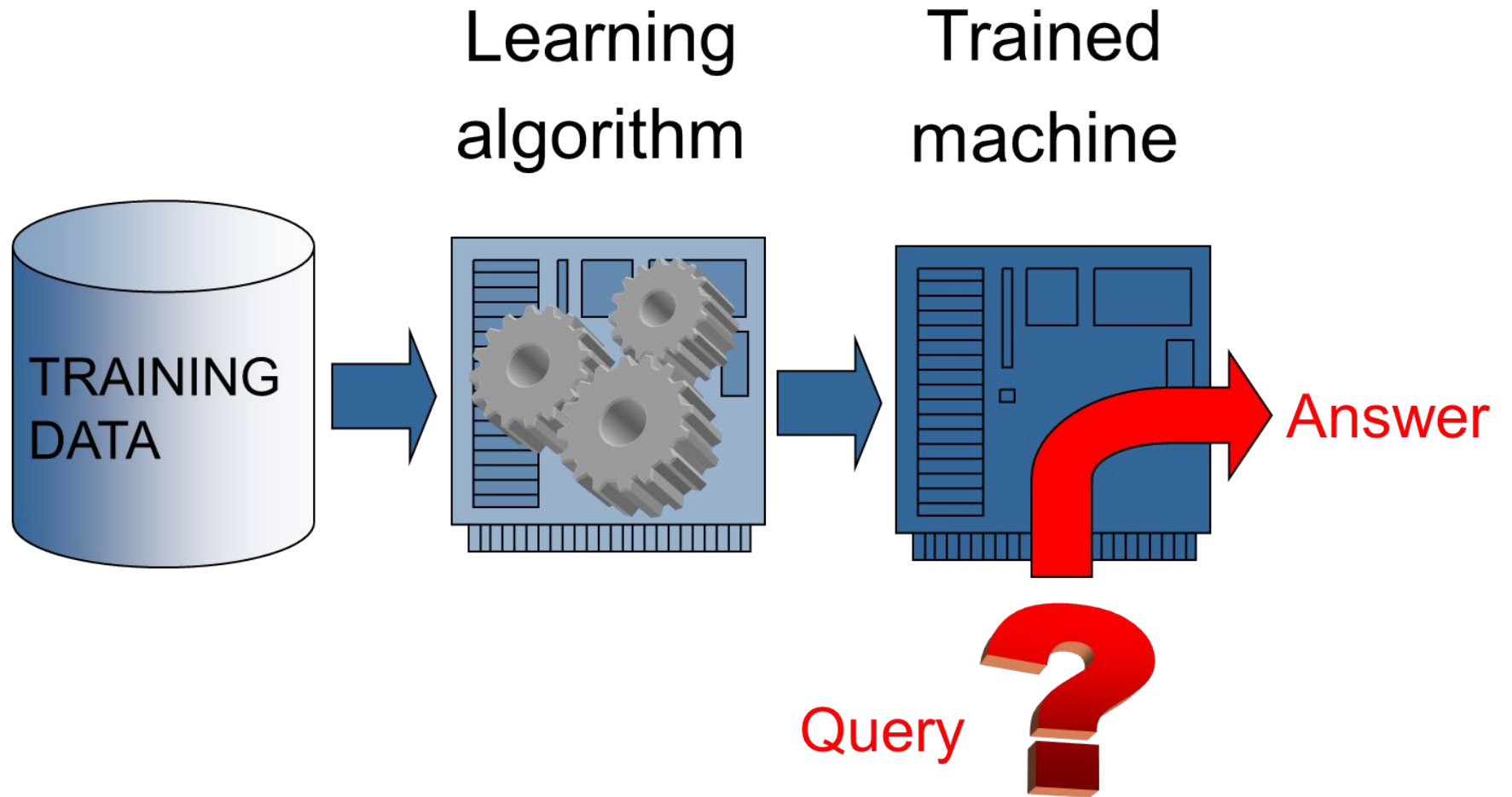


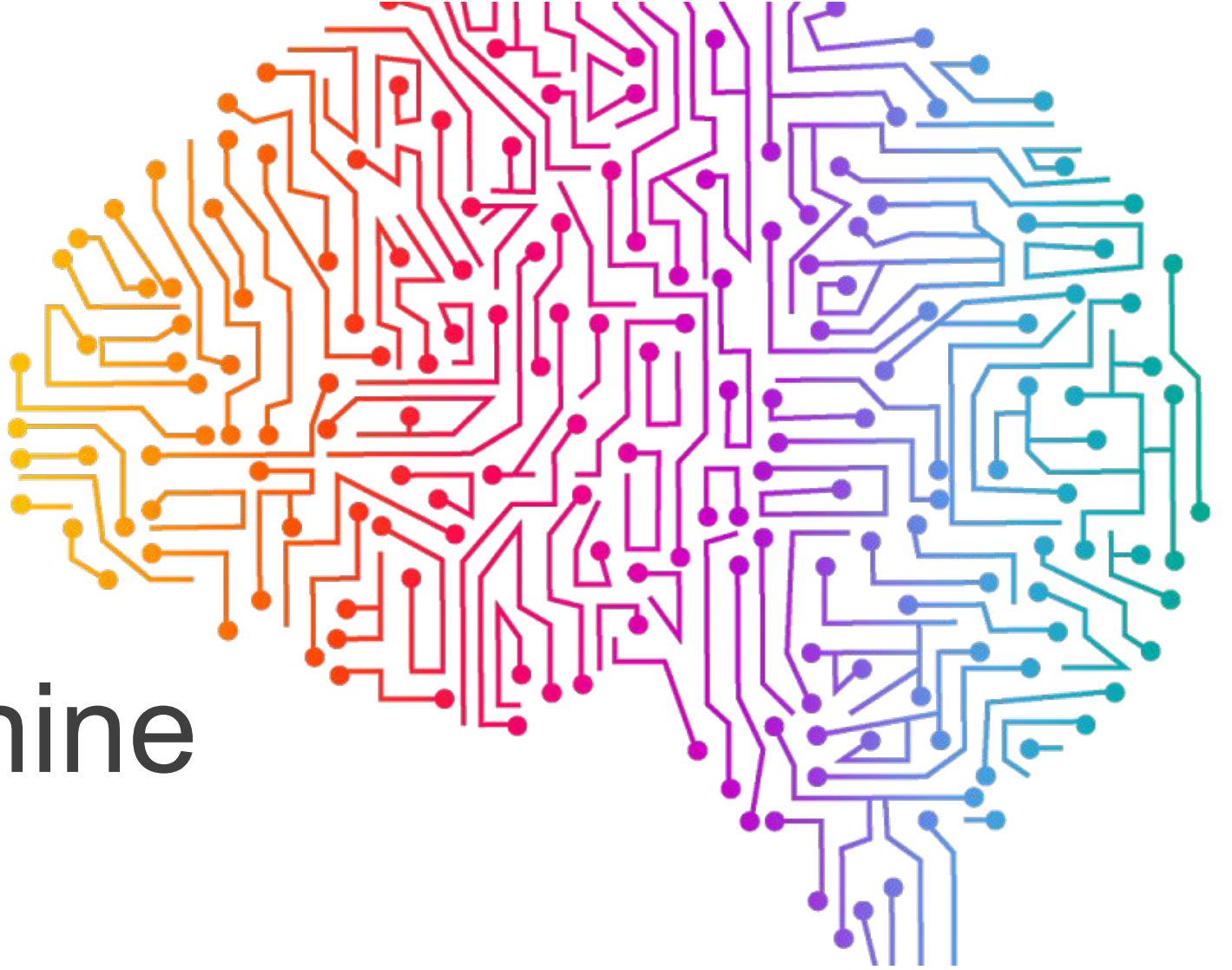
Machine Learning



- ❑ There is no need to “learn” to calculate payroll
- ❑ Learning is used in:
 - ❑ Data mining programs that learn to detect fraudulent credit card transactions
 - ❑ Programs that learn to filter spam email
 - ❑ Programs that learn to play checkers/chess
 - ❑ Autonomous vehicles that learn to drive on public highways
 - ❑ Self customizing programs
 - ❑ And many more...

Machine learning

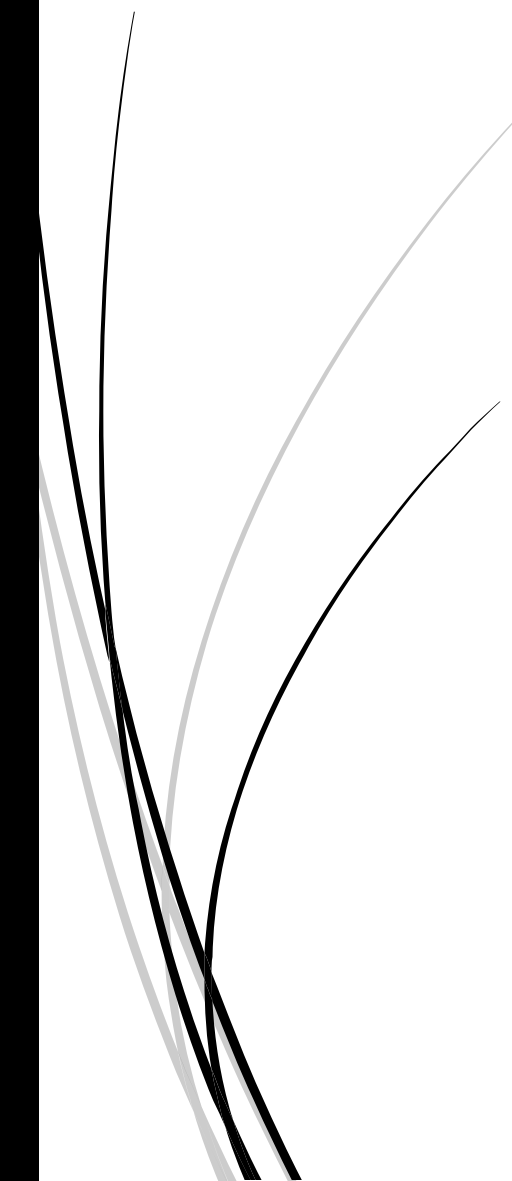




Types of Machine Learning

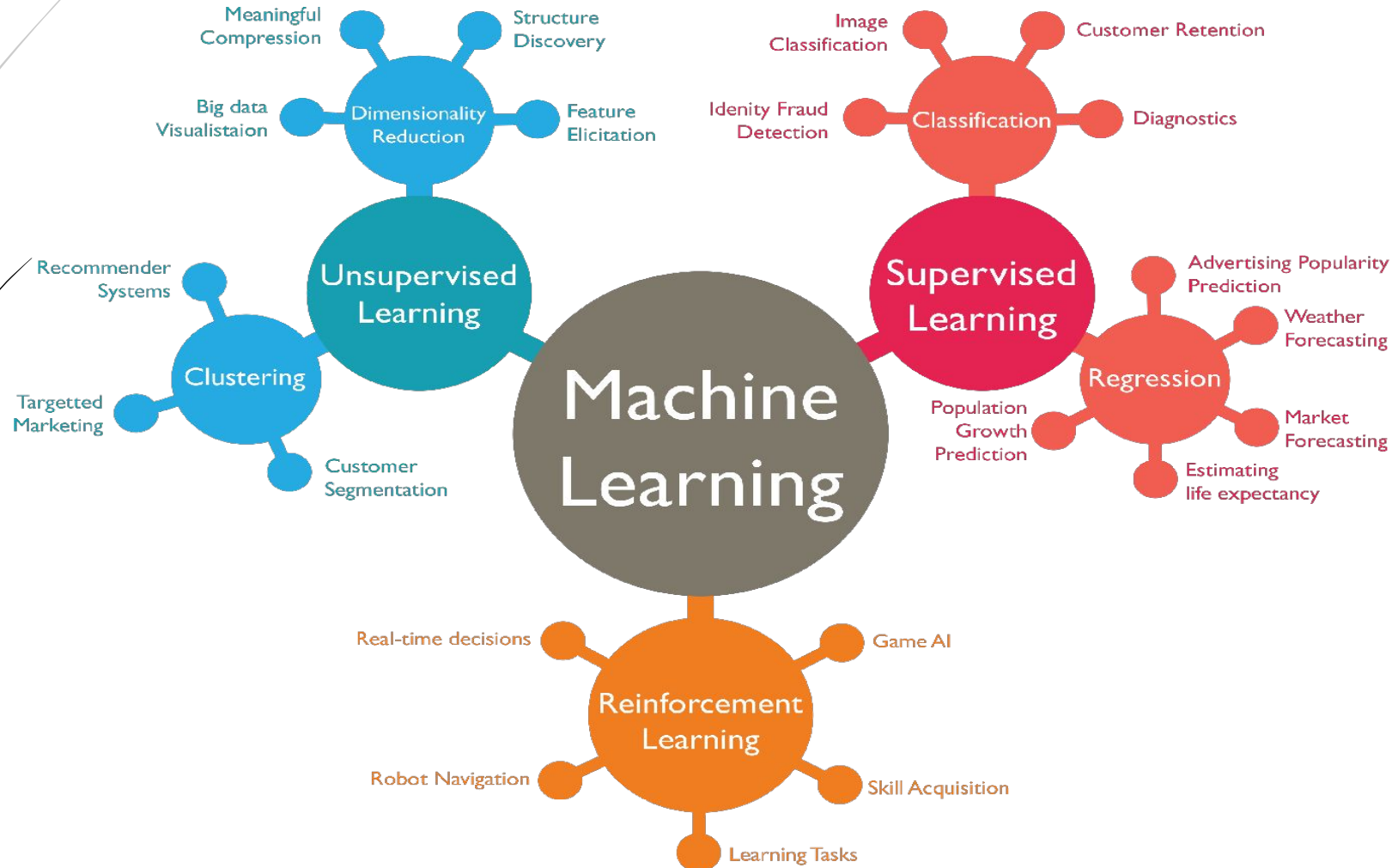


Types of Machine Learning



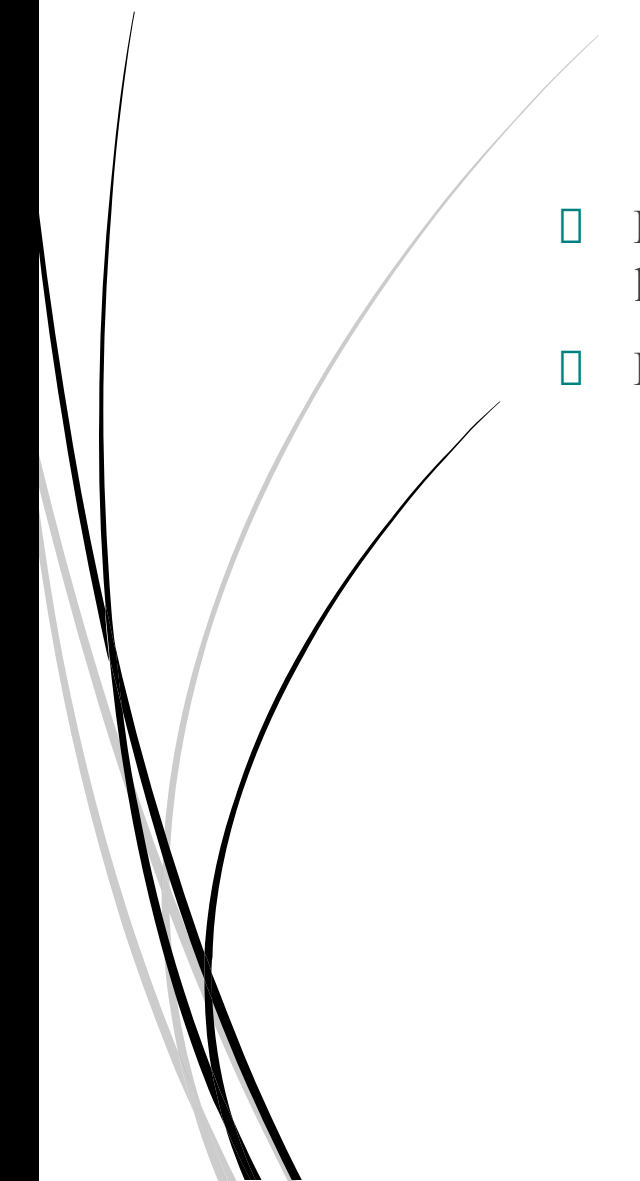
- – Supervised learning: learn mapping from attributes to “target”
Classification: target variable is discrete (e.g., spam email)
Regression: target variable is real-valued (e.g., stock market)
- – Unsupervised learning: no target variable; “understand” data structure
Clustering: grouping data into K groups
- – Other types of learning
 - Reinforcement learning: e.g., game-playing agent
 - Learning to rank, e.g., document ranking in Web search
 - And many others....

Machine Learning Types





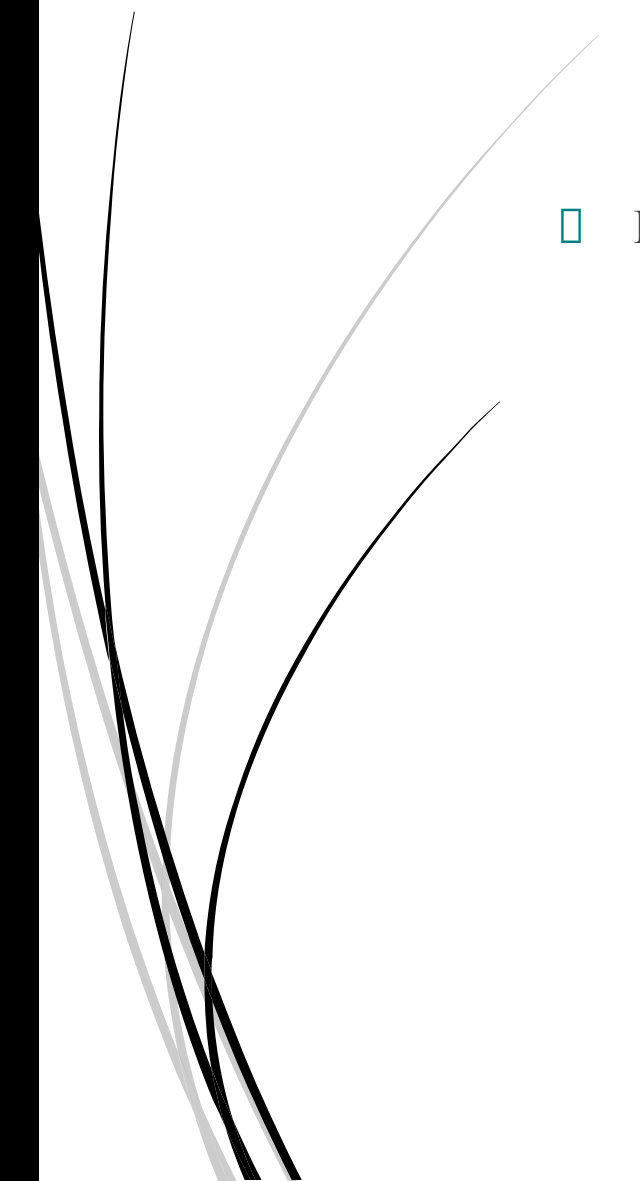
Supervised Learning



- ❑ Requires a labelled training set. A training set comprises of object belonging to different labels/classes.
- ❑ Example (face recognition):
 - ❑ Training Set: 10 persons or labels with 3 face images per person, each from different pose. We have 30 images in total. Labels in this case are names of each person (string).
 - ❑ Testing Phase: Take a random person from those 10, take his/her image from random pose and random facial expression (sad, smily etc) and ask which person that face belongs to.



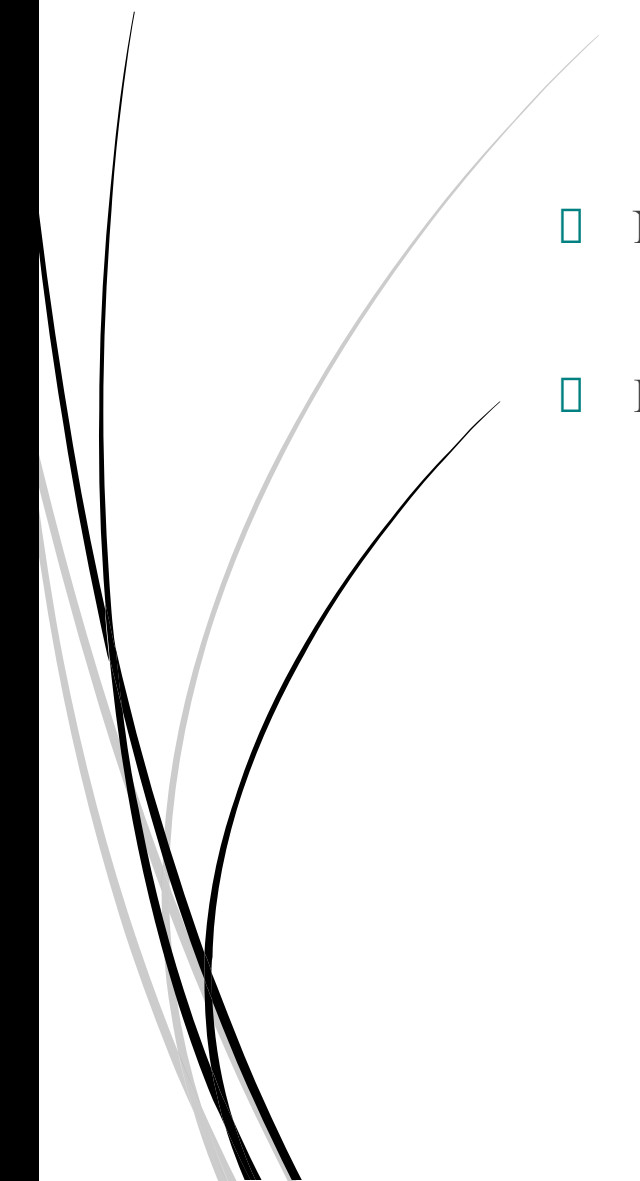
Supervised Learning



- Example (regression, estimating price of a plot):
 - Training Set: Imagine from pwd housing society, we have 10 plots and for each plot we know its covered area in square meters along with the price. Price in this case is the label (floating number)
 - Testing Phase: Estimate the price (label) of a random plot given its covered area.



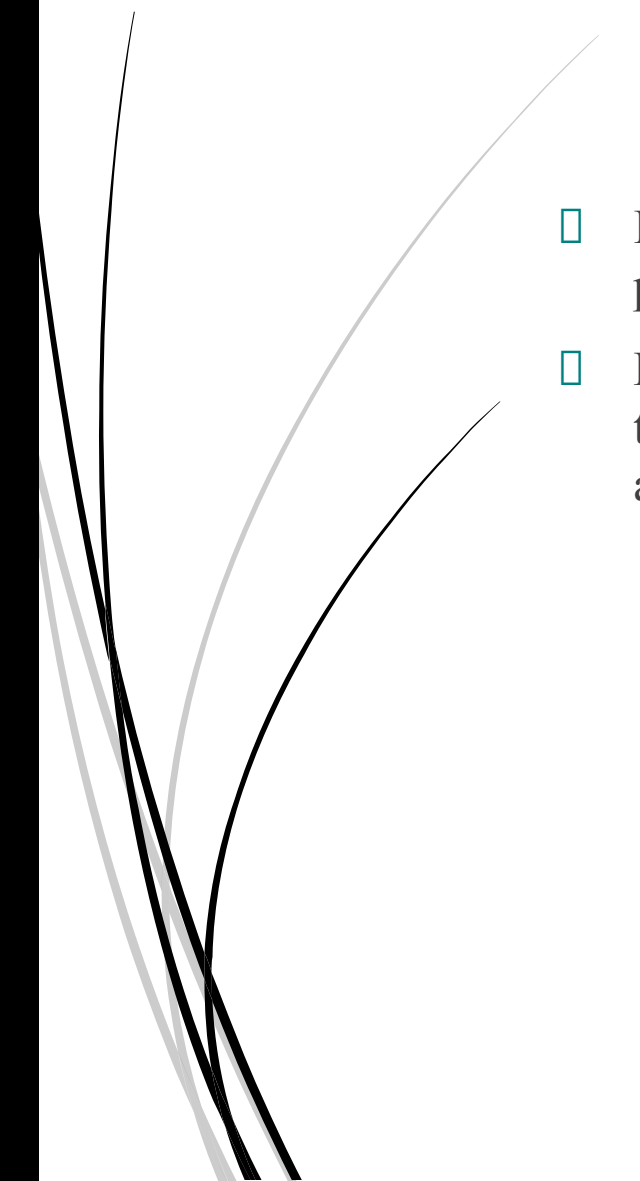
Unsupervised Learning



- No labelled training set required.
- Example (Clustering):
 - Imagine you have 50 students and for each student you know his/her marks out of 100. Divide these 50 students in good, average and worst categories based upon their marks.
 - Label in this case is one of the three strings (“good”, “average” or “bad”). Here you have a training set (a list of marks of those 50 students) but you don’t know the label of each of that student (whether that student belongs to a “good” category or one of the other two. That is what you have to find out!



Reinforcement Learning

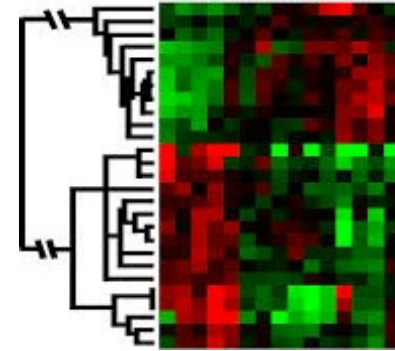


- In **reinforcement learning** the agent learns from a series of reinforcements—rewards or punishments.
- For example, the lack of a tip at the end of the journey gives the taxi agent an indication that it did something wrong. The two points for a win at the end of a chess game tells the agent it did something right.

When Do We Use ML?

ML is used when:

- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise (speech recognition)
- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)



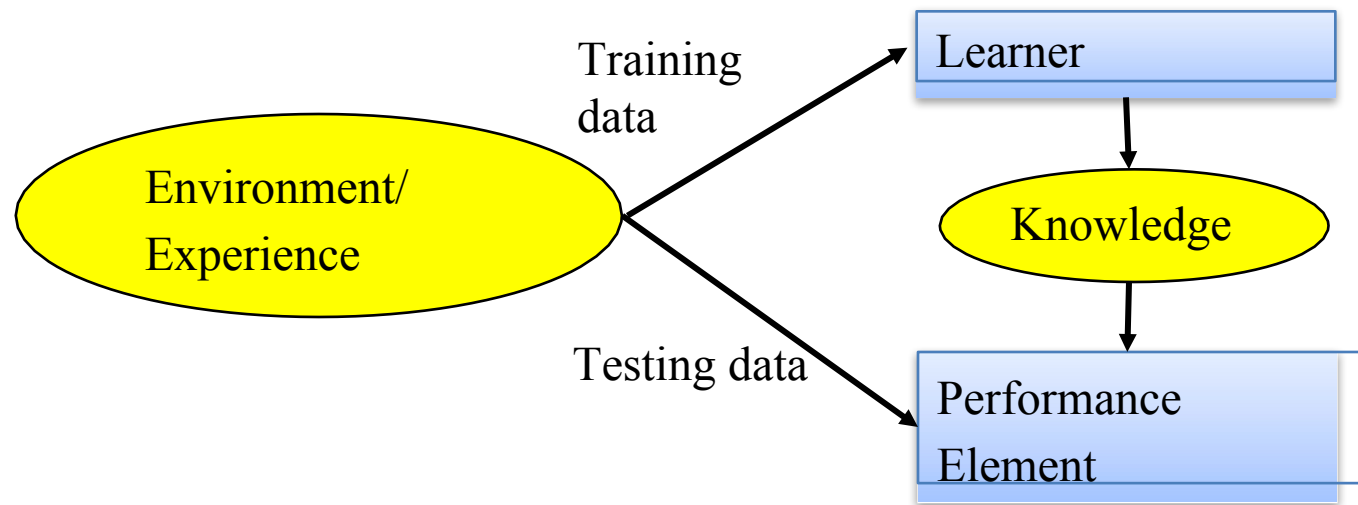
Learning isn't always useful:

- There is no need to “learn” to calculate payroll

Based on slide by E. Alpaydin

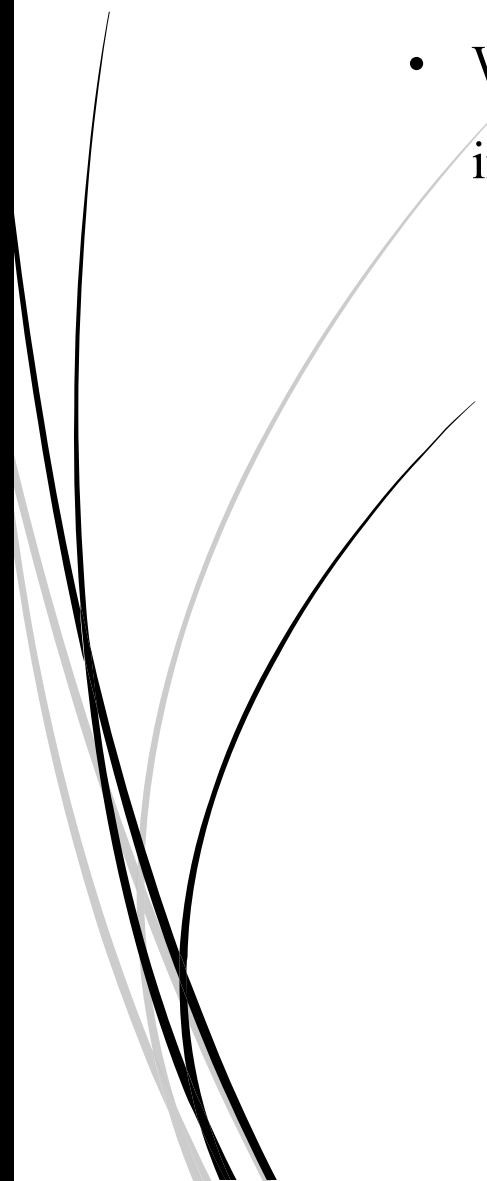
Designing a Learning System

- Choose the training experience
- Choose exactly what is to be learned
 - i.e. the *target function*
- Choose how to represent the target function
- Choose a learning algorithm to infer the target function from the experience



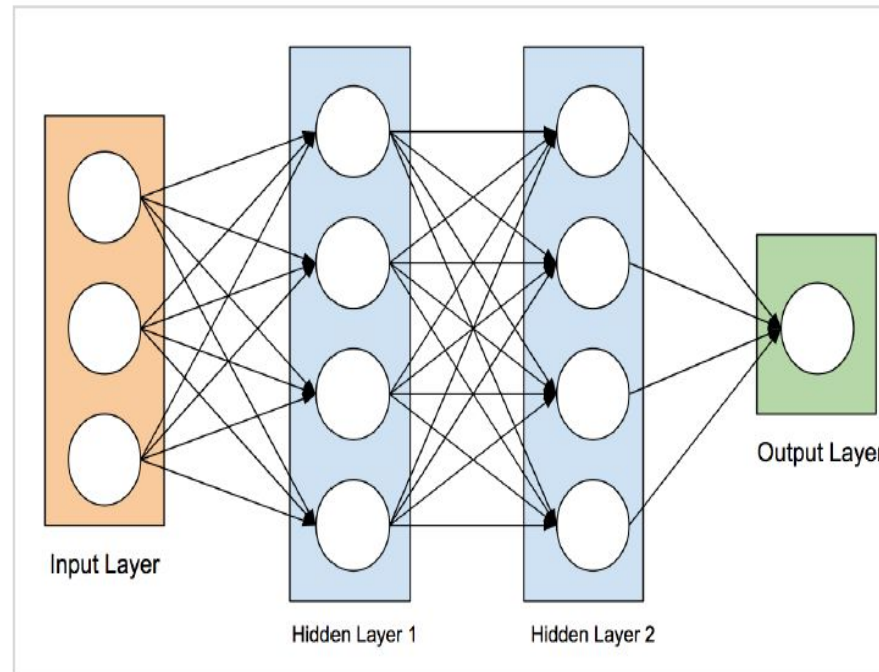


Training vs. Test Distribution

- We generally assume that the training and test examples are independently drawn from the same overall distribution of data
 - We call this “i.i.d” which stands for “independent and identically distributed”
 - If examples are not independent, requires *collective classification*
 - If test distribution is different, requires *transfer learning*
- 

Artificial Neural Network

The idea of artificial neural networks was derived from the neural networks in the human brain. The human brain is really complex. Carefully studying the brain, the scientists and engineers came up with an architecture that could fit in our digital world of binary computers. One such typical architecture is shown in the diagram below:



ANN Architectures

A mostly complete chart of Neural Networks

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- Backfed Input Cell
- Input Cell
- Noisy Input Cell
- Hidden Cell
- Probabilistic Hidden Cell
- Spiking Hidden Cell
- Output Cell
- Match Input Output Cell
- Recurrent Cell
- Memory Cell
- Different Memory Cell
- Kernel
- Convolution or Pool

Perceptron (P)



Feed Forward (FF)



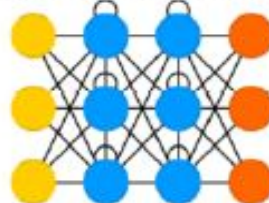
Radial Basis Network (RBF)



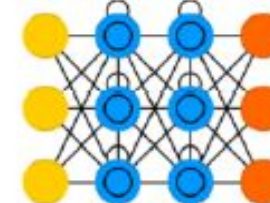
Deep Feed Forward (DFF)



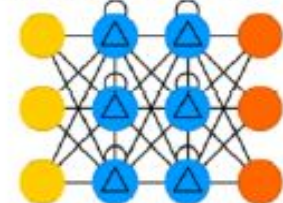
Recurrent Neural Network (RNN)



Long / Short Term Memory (LSTM)



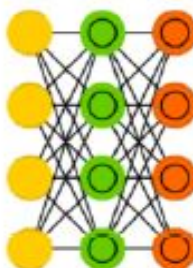
Gated Recurrent Unit (GRU)



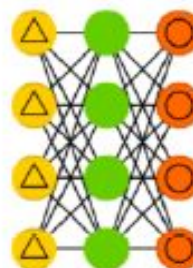
Auto Encoder (AE)



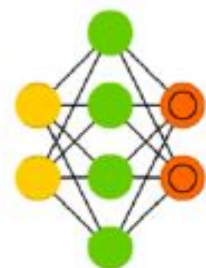
Variational AE (VAE)



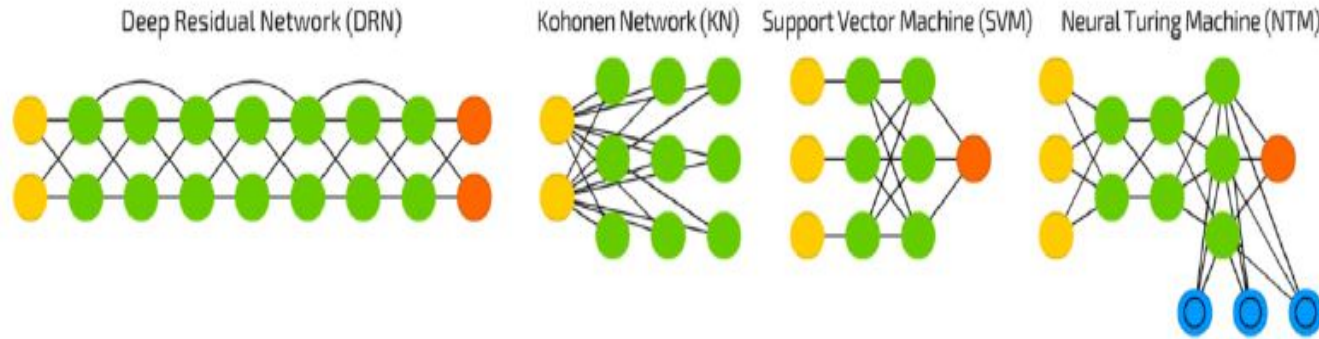
Denoising AE (DAE)



Sparse AE (SAE)



ANN Architectures



Each architecture is developed for a specific type of application. Thus, when you use a neural network for your machine learning application, you will have to use either one of the existing architecture or design your own. The type of application that you finally decide upon depends on your application needs. There is no single guideline that tells you to use a specific network architecture.

Source:

<https://towardsdatascience.com/the-mostly-complete-chart-of-neural-networks-explained-3fb6f2367464>

Deep Learning

- The deep learning is a model based on Artificial Neural Networks (ANN), more specifically Convolutional Neural Networks (CNN)s.
- There are several architectures used in deep learning such as deep neural networks, deep belief networks, recurrent neural networks, and convolutional neural networks.
- These networks have been successfully applied in solving the problems of computer vision, speech recognition, natural language processing, bioinformatics, drug design, medical image analysis, and games.
- The deep learning requires huge processing power and humongous data, which is generally easily available these days.

Deep Learning

- Deep Learning uses ANN. First we will look at a few deep learning applications that will give you an idea of its power.

Applications

Deep Learning has shown a lot of success in several areas of machine learning applications.

- **Self-driving Cars:** The autonomous self-driving cars use deep learning techniques. They generally adapt to the ever changing traffic situations and get better and better at driving over a period of time.
- **Speech Recognition:** Another interesting application of Deep Learning is speech recognition. All of us use several mobile apps today that are capable of recognizing our speech. Apple's Siri, Amazon's Alexa, Microsoft's Cortana and Google's Assistant – all these use deep learning techniques.
- **Mobile Apps:** We use several web-based and mobile apps for organizing our photos. Face detection, face ID, face tagging, identifying objects in an image – all these use deep learning.

Deep Learning

Untapped Opportunities of Deep Learning

- ❑ Agriculture is one such industry where people can apply deep learning techniques to improve the crop yield.
- ❑ Consumer finance is another area where machine learning can greatly help in providing early detection on frauds and analyzing customer's ability to pay.
- ❑ Deep learning techniques are also applied to the field of medicine to create new drugs and provide a personalized prescription to a patient.
- ❑ The possibilities are endless and one has to keep watching as the new ideas and developments pop up frequently.

Deep Learning

What is Required for Achieving More Using Deep Learning?

- ❑ To use deep learning, supercomputing power is a mandatory requirement. You need both memory as well as the CPU to develop deep learning models.
- ❑ Fortunately, today we have an easy availability of HPC – High Performance Computing.
- ❑ Due to this, the development of the deep learning applications that we mentioned above became a reality today and in the future too we can see the applications in those untapped areas that we discussed earlier.



Deep Learning

Deep Learning - Disadvantages

Some of the important points that you need to consider before using deep learning are listed below:

- ❑ Black Box approach
- ❑ Duration of Development
- ❑ Amount of Data
- ❑ Computationally Expensive

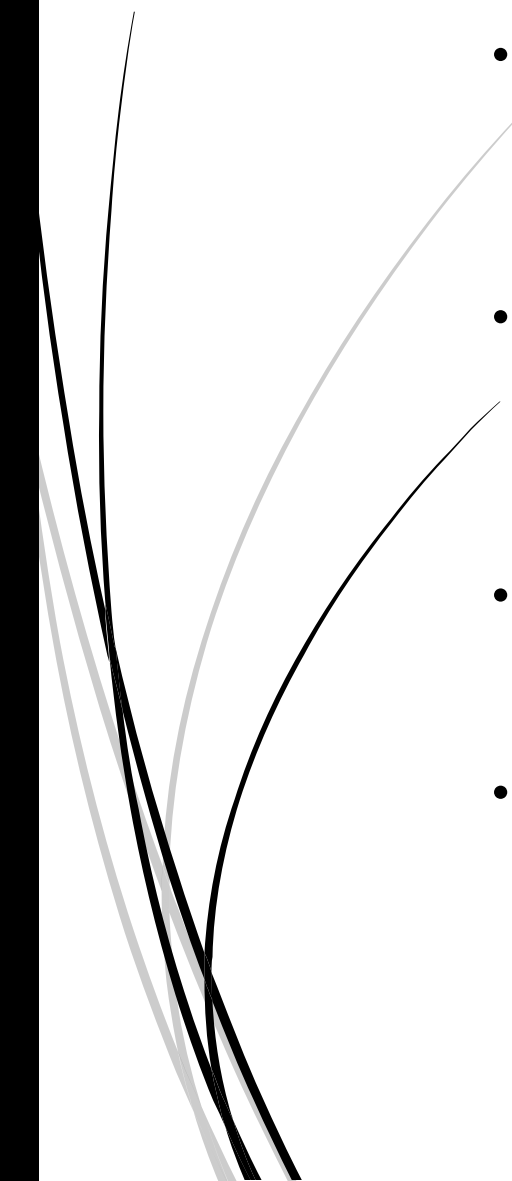


ML in a Nutshell

- Tens of thousands of machine learning algorithms
 - Hundreds new every year
- Every ML algorithm has three components:
 - **Representation**
 - **Optimization**
 - **Evaluation**

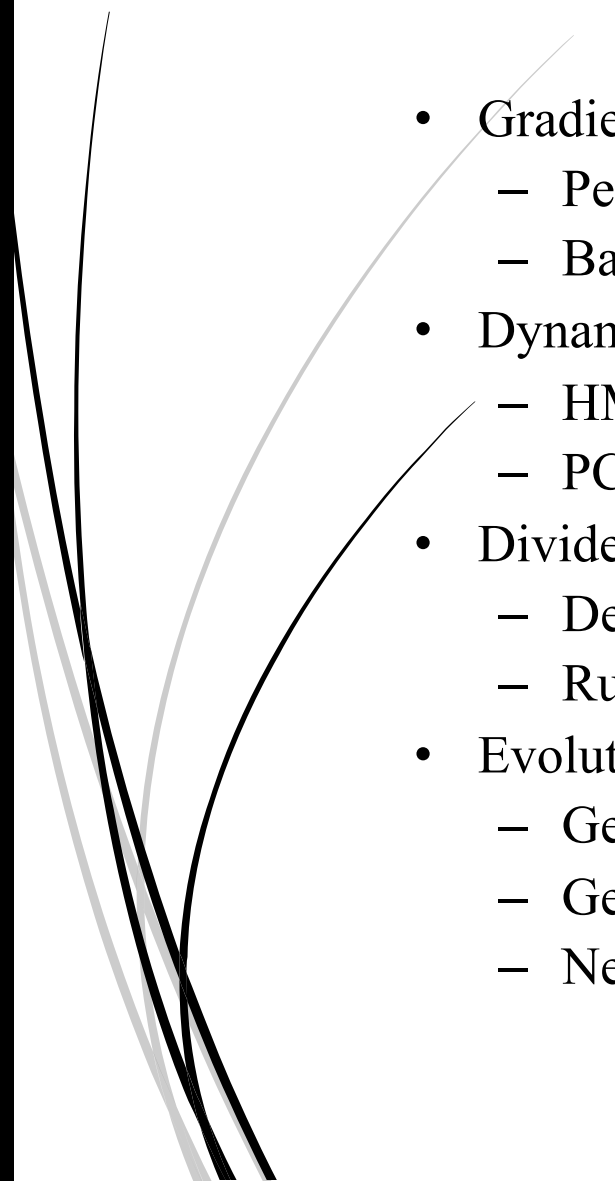


Various Function Representations

- Numerical functions
 - Linear regression
 - Neural networks
 - Support vector machines
 - Symbolic functions
 - Decision trees
 - Rules in propositional logic
 - Rules in first-order predicate logic
 - Instance-based functions
 - Nearest-neighbor
 - Case-based
 - Probabilistic Graphical Models
 - Naïve Bayes
 - Bayesian networks
 - Hidden-Markov Models (HMMs)
 - Probabilistic Context Free Grammars (PCFGs)
- 



Various Search/Optimization Algorithms

- Gradient descent
 - Perceptron
 - Backpropagation
 - Dynamic Programming
 - HMM Learning
 - PCFG Learning
 - Divide and Conquer
 - Decision tree induction
 - Rule learning
 - Evolutionary Computation
 - Genetic Algorithms (GAs)
 - Genetic Programming (GP)
 - Neuro-evolution
- 

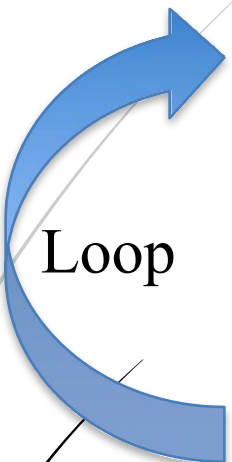


Evaluation

- Accuracy
- Precision and recall
- Squared error
- Likelihood
- Posterior probability
- Cost / Utility
- Margin
- Entropy
- K-L divergence
- etc.



ML in Practice



Loop

- Understand domain, prior knowledge, and goals
- Data integration, selection, cleaning, pre-processing, etc.
- Learn models
- Interpret results
- Consolidate and deploy discovered knowledge

Conclusion

- ❑ ML is a technique of training machines to perform the activities a human brain can do, albeit bit faster and better than an average human-being.
- ❑ Today we have seen that the machines can beat human champions in games such as Chess, AlphaGO, which are considered very complex. You have seen that machines can be trained to perform human activities in several areas and can aid humans in living better lives.
- ❑ Machine Learning can be a Supervised or Unsupervised. If you have lesser amount of data and clearly labelled data for training, opt for Supervised Learning.
- ❑ Unsupervised Learning would generally give better performance and results for large data sets. If you have a huge data set easily available, go for deep learning techniques.

